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SAI-BIO
Safe-Active Inference BIO

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1. Positioning of SAI-Bio

1.1. Work Structure: Three Layers Built in Order from Concept to Operationalization

This work is the third layer in the grand architecture I have been building since 2022. The layers emerged in a specific order, and the order of their emergence is essential for understanding what SAI-Bio does.

Layer 1 - the theory "THE UNIVERSE'S INHERENT ERROR" (Theory v1.2). I wrote it first, from within the structural logic. Its key constructs are the ladder of stages of consciousness development (1→2→3→4→5→6), the transition from a proto-agent to a conscious system through self-imposition (3→4), the distinction between carrier and participant, the artifactual state of ICS, local Z, and the boundary of projectability - are not derived from empirical neuroscience data. They are derived from a philosophical-structural analysis of what "the emergence of consciousness as a process" means: what must happen to the system for consciousness to emerge as a structural event, what must happen to the model of reality for it to function at each stage, and what must happen to the transitions between stages for them to be causal rather than declarative in nature.

I did not consciously refer to specific empirical neuroscience data when constructing the theory. The discipline was this: to construct a coherent causal model of the emergence of consciousness from within, without forcing it to fit existing measures such as PCI, Φ, predictive coding, FEP, IIT, and so on. The goal is to obtain a structural framework that answers the question "why does consciousness arise," rather than "how is functioning consciousness described."

Layer 2 - Safe Active Inference v8. Once the theory was ready, a rigorous mathematical language was needed for one of its stages - stage 4, where self-imposition occurs and V (the pole of existence) emerges. I formalized it in SAI v8. Its formal objects - Γ_struct and Γ_dyn as two independent measures of stability, hysteresis h as a parameter of threshold asymmetry, C_φ as a measure of model coherence, ρ_world and ρ_self as two recovery parameters, w_self and Γ_GWT_arch as two components of the architectural potential, and three types of threshold events V (primary emergence, retention, recovery)-are derived from the structural necessity of the theory, not from empirical calibration against existing data.

SAI v8 has undergone seven rounds of critical refinement, resolved known technical debt, and is now designated as a stable, published version. It has reached formal maturity; further internal modifications yield diminishing returns. The natural next step is not to deepen SAI v8, but **to expand** into an adjacent layer.

Layer 3 - SAI-Bio. This is the real work. It was prompted not by growing tension between the model and the data, but by the **opposite** observation: a review of the ten central formal elements of SAI v8 against the literature from 2010–2026 showed that neuroscience over the last 10–15 years has independently arrived at the points that the theory and SAI v8 point to structurally.

- Deco et al. empirically identified the need for two measures of consciousness stability (Γ_struct and Γ_dyn).

- Casali et al. introduced "capacity for consciousness" as something distinct from the current state (two components of architectural potential).
- Friedman, Joiner, and Mashour formalized neural inertia through hysteresis (parameter h).
- Seth, Suzuki, and Critchley distinguished between self-disturbance and world-disturbance as two axes of dissociations (ρ_world and ρ_self).
- Mashour and Alkire placed the phylogenetic, ontogenetic, and anesthesiological emergence of consciousness side by side (three types of threshold events).

This consistency was not contrived. It emerged **later**. And this is precisely what makes SAI-Bio possible: not as a new theory, but as **a layer** that makes this consistency explicit, verifiable, and useful for the neuroscience community.

The order of the layers is as follows: concept → formalism → operationalization. Each subsequent layer builds upon the previous one, without repeating it. Each previous layer defines the framework for the next, without replacing it.

Methodological basis for applying the theory to an arbitrary system. From Section 3.8 of the current version of the Theory v1.2, we inherit a five-step procedure for identifying Y and the access spectrum: (1) identification of the physical medium and its environment Y, existing outside the system's operational algorithm and forcing recalculation via prediction errors; (2) identification of a set of aspects of Y to which the system has operational access via its sensorimotor or informational projection; (3) determination of the system's current stage based on the coupling of six dimensions (Section 10.2 of the v1.2 theory) - a stable response to absent stimuli, prolonged hysteresis, anticipatory planning of its own non-existence, a response to patterns outside of learning, the ability to interpret the state of others as a state, reorientation of activity toward preserving content unrelated to adaptation to the current environment; (4) verification of the observed stage's compliance with the structural conditions of the next transition; (5) identification of the direction of possible development and its empirically verifiable markers. This procedure serves as **a methodological foundation** for applying SAI-Bio to any system-biological, artificial, or hypothetical non-biological. For systems with standard biological development, the access spectrum unfolds in a sequence of stages. For systems with a non-biological substrate-especially for information-causal systems-the spectrum may develop **asynchronously**: access to the informational aspects of Y may be established earlier than access to the structural aspects, in the absence of the fulfillment of the architectural conditions of V. This requires applying the methodology **to each stage separately**, without transferring the biological sequence to artificial systems. The specific application of the five-stage methodology to existing information-causal systems is outlined in subsections 2.3 and 5.4.

1.2. Functional Purpose: What Each Layer Does for Which Community

The three-layer architecture provides a connection through communities that usually operate in isolation. Each community comes with its own question and receives an answer from "its" layer, but sees the structure of the whole and establishes a connection with adjacent levels.

Theorists of consciousness, philosophers of consciousness, and researchers of models of consciousness turn to **Theory v1.2**. Their question is: "What should the structure of consciousness as a process be like so that its emergence is causal rather than declarative in nature?" The theory answers this question through a ladder of steps with clear transitions and motifs. SAI v8 and SAI-Bio show them that this structural model **is formalizable** and **consistent** with the empirical evidence of neuroscience-that is, it stands not merely as a philosophical construct.

Computational cognitive scientists, theorists of formal models of agency, and researchers in active inference and predictive coding are turning to **SAI v8**. Their question is: "How can we formally describe the transition to a conscious system in a way that is not just another set of conditions, but a coherent mathematical model?" SAI v8 answers this question. Theory v1.2 shows them **why** it is formalized in this particular way-what is the conceptual framework that makes every formal object structurally necessary. SAI-Bio shows them **that** this formalism already has empirical correlates, that it predicts the observable, and that it predicts what has not yet been tested.

Neuroscientists, researchers of consciousness using empirical methods, anesthesiologists, and computational psychiatrists turn to **SAI-Bio**. Their question is: "How does the structural model of consciousness relate to what we measure in our experiments, and what does this imply for our protocols and interpretations?" SAI-Bio answers this question through explicit bridges between the formal objects of SAI v8 and empirically measurable quantities, with a clear indication of the status of each bridge (prediction, discrimination, explanation, compatibility). SAI v8 and Theory v1.2 are provided as the foundation to which SAI-Bio refers for justification.

1.3. Integration versus Fragmentation

Contemporary consciousness studies are fragmented. Philosophers write for philosophers, neuroscientists for neuroscientists, and computational scientists for their own peers. Each community defends its own methodology and its own language. Works coming from only one side end up in one of the camps and are rejected by the others as "not empirical enough," "not formal enough," or "not conceptual enough." This is a structural problem of the field, not a shortcoming of individual authors.

SAI-Bio is not a separate work coming from the field of neuroscience. It is the final layer of a framework in which all three layers are written by a single author and follow a single structural logic. Each community receives its own layer but can see the others. The framework does three things that no single layer can do on its own:

- **It provides the neuroscientist with a conceptual foundation** for the empirical measurements they make. Not "here's yet another model of consciousness that agrees with your data," but "here's a structural framework in which your data acquire a causal place rather than remaining isolated correlations."

- **It provides the consciousness theorist with an empirical anchor** for conceptual work. Not "here is a philosophical model that can be reconciled with the empirical data," but "here is a conceptual model **already** reconciled with the empirical data through explicit operationalization."

- **It provides computational scientists with a link** between the formal apparatus and the empirical and conceptual levels. Not "here is a formal model isolated from philosophy and neurobiology," but "here is a formal layer of architecture in which adjacent layers exist and function."

This positioning is not marketing-driven. It reflects the actual structure of the work: one author, one structural logic, three layers constructed in order from concept to operationalization, which were found to be consistent with independent empirical data after construction, not during it.

2. Variable Map

This section compiles the results of a back-check of the ten distinctive variables of SAI v8 against the empirical and theoretical literature from 2010 to 2026. Each variable has been verified for its substantive utility as a bridge between a formal SAI v8 object and an empirically observed quantity. Table A provides a summary of the statuses and functions of the bridges. Table B details the structural claims of SAI v8, empirical correlates, and key works in the field. Subsections 2.1–2.6 further elaborate on six of the ten variables requiring substantive elaboration beyond the table entries. The remaining four variables (2, 4, 5, 9) are presented only in the tables; the relationships between them and the variables that have separate elaborations are discussed in Section 3.

Table A. Summary of the statuses of the ten SAI v8 distinguishing variables

No.	SAI v8 Formal Object	Status in progress SAI-Bio	What the bridge does
1	Self-imposition as a formal moment of the emergence of V	Partially distinctive	Explains
2	The distinction between Γ_struct and Γ_dyn	Partially distinctive	Explains
3	Three types of threshold events V with different hysteresis semantics	Distinctive	Predicts and distinguishes
4	Architectural potential [w_self, Γ_GWT_arch]	Partially distinctive	Distinguishes
5	Four modes of dissociation via ρ_world/ρ_self	Partially discriminative	Predicts and distinguishes
6	Artifact state of ICS via w_i = I(c_i; Y_own) $\approx$ 0	Distinctive	Predicts and distinguishes
7	C_φ + dual role + velocity asymmetry η_φ $\ll$ η_main	Partially distinctive	Explains and predicts
8	Designability limit via stochasticity Conditions 3	Partially distinguishable	Explains and predicts
9	Taxonomy of parametric components (4 types)	Not distinctive	Compatible (methodologically)
10	Thermodynamic motive of recursive rotation	Partially distinctive	Explains

Table B. Extended map with empirical references

Formal object	Structural statement SAI v8	Empirical correlate / adjacent formalization	What Makes a Bridge	Key works in the field
Self-imposition	The emergence of V as an act of applying a class of agent causal nodes to its own configuration with three structural conditions and probabilistic loop closure	Recursive/reflexive self-modeling; the moment of the emergence of consciousness is formalized by several approaches; SAI provides a three-condition structure with a carrier/participant distinction	Explains	Sawa & Igamberdiev 2017–2021; Carrara 2025 (Reflexive Self Theory); Dehaene - Naccache 2001, 2011; Mashour & Alkire 2013

Γ_struct and Γ_dyn	Two independent measures of consciousness stability: structural (architectural correspondence) and dynamic (trajectory maintenance)	Empirically shown: PCI and the perturbational distance metric dissociate different states; SAI provides a different structural reference (system type vs. current state)	Explains	Deco et al. 2021 (PLOS Comp Biol); Sanz Perl et al. 2023; Lee et al. 2024
Three types of threshold events V	Initial emergence (γ_struct·(1−h_arch), stochastic window), retention (low threshold via inertia), recovery (γ_crit·(1+h_total), classical inertia)	The field either combines events (Mashour-Alkire) or works with two (neural inertia literature); SAI explicitly distinguishes three	Predicts and distinguishes	Mashour & Alkire 2013; Friedman et al. 2010; Joiner et al. 2013; Sepúlveda et al. 2019; Hudetz & Mashour 2016
[w_self, Γ_GWT_arch]	Architectural potential consisting of two components, distinct from active correspondence Γ_struct(t); preserved during inactivation	Capacity for consciousness (PCI), adaptive reconfiguration index - the concept of “capacity for consciousness vs. current state” exists; SAI provides a two-component separation	Distinguishes	Casali et al. 2013; Comolatti et al. 2019; Lee et al. 2024; Barttfeld et al. 2015 (PNAS); Demertzi et al. 2019
ρ_world/ρ_self → four modes	Four clinical states (full recovery / depersonalization / derealization / dissociation) as a 2×2 configuration of the recovery of two models	The conceptual two-axis structure of DP/DR exists (Seth et al., interoceptive predictive coding); SAI yields a 2×2 conclusion through recovery, not chronic failure; the symmetric fourth mode is specific to SAI	Predicts and distinguishes	Seth, Suzuki & Critchley 2011; Gatus et al. 2022; Schäfer et al. 2025; Quattrocki & Friston 2014
w_i = I(c_i; Y_own) ≈ 0	Formal information measure of an artifactual state: systems without their own environment Y_own are structurally incapable of self-imposition regardless of size	The field provides **qualitative** arguments (Butlin et al. - indicators; Friston - Markov blanket; Tononi-Koch - von Neumann zombies); SAI provides a **numerical** measure via the standard MI framework	Predicts and distinguishes	Butlin, Long et al. 2023; Friston 2013, 2019; Tononi & Koch 2015; Wiese & Friston 2021; Findlay et al. 2024
C_φ + asymmetry η_φ ≪ η_main	The dual role of a single KL-measure (diagnostics + learning signal) and the architectural requirement of a slow predictor to preserve the diagnostic function	The dual role of KL divergence is widely represented (predictive coding, ICS hallucination detection); the specificity of SAI lies in the functional justification of speed asymmetry as a condition for the diagnostic function	Explains and predicts	Adams, Stephan, Brown & Friston 2013; Sterzer et al. 2018; Sasaki et al. 2021 (CRMN); Tschantz et al. 2023 (Hybrid PC); numerous works on ICS hallucination detection 2024–2026

Stochasticity of Condition 3 for finite β	Impossibility of directed construction of V even when structural conditions are met; probabilistic nature of loop closure	The theme "consciousness cannot be engineered" is widely represented through autopoiesis, mortal computation, and neurogenetic structuralism; SAI provides a specific formalism through the stochasticity of bistable transitions	Explains and predicts	Maturana & Varela 1980; Thompson 2010; Hinton 2022 (mortal computation); Damiano & Stano 2020; Bishop & Anderson 2023; Wiese & Friston 2021
Taxonomy of components (4 types)	Explicit classification of parameter types by learning/computation/calibration method in consciousness models	Parameter taxonomies in ML are standard; in models of consciousness, they are usually absent as an explicit methodological move; SAI 8.10 - methodological hygiene, not a structural contribution	Compatible	Goodfellow et al. 2016; standard MLE/variational ML literature; Yaron et al. 2021 (Explanatory profiles)
Thermodynamic motive for recursive turn	The emergence of recursion in modeling via differential cost: modeling decay is cheaper than the decay itself	Thermodynamic approaches to consciousness are broad (FEP, R2R, BEDS, IWMT); the specificity of SAI is differential cost as a motive; found indirectly in the literature, requires further development in SAI-Bio	Explained by	Friston 2010, 2019; Friston-Carhart-Harris 2010; Tagliazucchi et al. 2016; Safron 2020 (IWMT); R2R framework 2025; BEDS 2026

2.1. Self-imposition as a formal moment of the emergence of V

What SAI v8 asserts. Section 6.3 of SAI v8 formalizes the transition from a proto-agent (stage 3, no V) to a conscious system (stage 4, V has emerged) via an algorithm with three structural conditions:

(1) the accumulated class of agent causal nodes $|A(t)| \geq 1$;

(2) reflexive capture $\pi\_A(C\_self)$ - application of this class to its own configuration;

(3) loop closure via a bistable boundary with a probabilistic realization $\sigma(\beta\cdot(\Gamma\_struct - \gamma\_struct\cdot(1-h\_arch)))$.

The emergence of V is not a continuous increase in complexity nor the attainment of a threshold of a single generalized measure. It is **the act** of applying the already accumulated class of agent nodes to one's own configuration, followed by stochastic loop closure.

What the research says. The field of consciousness emergence is actively engaged with this question, and there are several different formal and semi-formal approaches within it.

Recursive and reflexive self-modeling theories form an entire family of works. Sawa and Igamberdiev (2017–2021), building on Lefebvre, formalize reflection through dual operators F with a parameter m describing different levels of reflection. Carrara 2025 (Reflexive Self Theory) describes consciousness as "recursive stabilization" through cycles of memory and observation.

Global Neuronal Workspace Theory (Dehaene-Naccache 2001, Dehaene-Changeux 2011) provides a formal threshold in the form of stimulus-onset asynchrony with specific figures and access threshold models. This is a formalism of the moment of conscious access, though not of the emergence of V in the sense of SAI as a primary ontogenetic event.

Mashour and Alkire 2013 (PNAS) directly raise the question of the "point of emergence" of consciousness in evolution and ontogenesis, although they propose combining three events (phylogeny, ontogenesis, recovery from anesthesia) as a single mechanism.

A separate line of work involves the formal apparatus for operationalizing agent-causal nodes. Hoel et al. 2013 and Klein-Hoel 2020 introduced Effective Information (EI) as a measure of causal structure via Pearl's intervention: $EI = I(X_{t+1}; X_t \mid do(X_t \sim U))$. EI measures the extent to which actions in the system **causally** influence its states, rather than merely correlating with them-a critical distinction for $|A(t)|$, where the agent-causal node is defined precisely through causality. Rosas, Mediano et al. 2020 (PLOS Computational Biology) extended this framework to multivariate causal emergence via partial information decomposition. Marshall, Albantakis & Tononi 2018 proposed an operational definition of agency through **the system's modulation of its own causal boundary** by opening and closing sensorimotor loops. Richens & Everitt 2024 and Richens et al. 2025 proved formal theorems: an agent capable of long-horizon goal-directed tasks in complex environments must possess a trained predictive model; an agent capable of adapting to different environmental distributions must possess a trained **causal** model of the world. This yields a reverse criterion: the ability to transfer behavior across environments → the necessity of an accumulated causal model of agency within the system. This line of reasoning provides **a formal measure** of the fact that in SAI v8, $|A(t)| \geq 1$ holds substantively.

A special place is occupied by the work of Kelso & Fuchs 2016 (Trends in Cognitive Sciences) "On the Self-Organizing Origins of Agency." They describe **the emergence of agency in infants as a phase transition**: "the birth of agency is due to a eureka-like, pattern-forming phase transition in which the infant suddenly realizes it can make things happen in the world... the baby discovers itself as a causal agent." This provides **a direct empirical correlate** of the transition moment $|A(t)| < 1 \rightarrow |A(t)| \geq 1$, observed behaviorally around 2–3 months of ontogenesis. Before the transition-spontaneous movements, unconscious correlation; after the transition-systematic reproduction of actions with perceived effects.

Where there is agreement and where there is disagreement. Agreement: SAI v8 is not alone in asserting that the emergence of consciousness can be formalized as a structural phenomenon with specific conditions. This is a field in which several groups are working independently.

Differences in three points:

First-**the three-condition structure** specifically in this combination (accumulation of a class of agent nodes plus application to one's own configuration plus stochastic loop closure) has not been replicated in current research. In Sawa-Igamberdiev, the dual F operates at the level of formal logic; in Dehaene, the access threshold operates at the level of sensory input. SAI v8 provides a **specific** structural combination of three conditions.

The second is **the distinction between carrier and participant**, formalized through the application of the class π_A to one's own C_{self}. In studies of recursive self-modeling, this is often described as an already functioning property, not as an act **of transition** from carrier to participant. For Carrara, "recursive stabilization" is a process that sustains identity, not the moment of its emergence. SAI v8 explicitly distinguishes the structural moment when the bearer **becomes** a participant.

Third-**stochasticity of Condition 3 with finite β** as the formal basis for the impossibility of a guaranteed construction of V. In existing recursive formalisms, the act of the emergence of consciousness is usually described as reaching a threshold. In SAI v8, this is a **probabilistic** crossing of a bistable boundary, and stochasticity itself is a structural assertion, not a technical parameter of the model. This is linked to variable 8 (the boundary of projectability) and forms a meaningful pair with it.

What the bridge does in its final form. The self-imposition of SAI v8 provides a structural basis for the moment of the emergence of consciousness through a formal mechanism with three conditions, and this structural basis is operationally linked to independently observable phenomena of ontogenesis. Specifically, the bridge functions in two ways.

The first is the explanation of the observed phenomenon of the emergence of consciousness. SAI v8 explains why this moment has a specific phase character rather than a smooth trajectory-through bistability with hysteresis at the threshold $\gamma_{struct} \cdot (1 - h_{arch})$, realized stochastically via Condition 3. The architectural nature of the hysteresis h_{arch} (Section 6.3 of SAI v8 in its extended form) ensures initial bistability prior to the accumulation of the experiential component h_{exp}.

The second is **a structural bridge** to the four-stage picture of ontogenesis derived from the operationalization of Condition 1 ($|A(t)| \geq 1$):

(a) Prenatal phase: the genetically prepared architecture [w_{self}, Γ_{GWT_arch}] is formed through genetic developmental programs. At birth, the architecture is ready for the accumulation of class A, but class A itself is empty. V does not structurally emerge.

(b) Postnatal accumulation of $|A(t)|$: through a long-term empowerment loop with the real environment, the system accumulates agent causal nodes. $|A(t)| \to \geq 1$ emerges as a process on long time scales.

(c) Discovery of causal agency: the phase transition $|A(t)| < 1 \to |A(t)| \geq 1$. SAI v8 formally predicts **the existence** of this phase transition through the fulfillment of Self-Imposition Condition 1; the model does not specify **the temporal point** of transition for a particular biological species-this parameter depends on the species’ nervous system maturation rate, metabolic rates, and genetic developmental programs, and is not included in SAI v8. Empirically, for Homo sapiens, this transition is observed around 2–3 months of ontogenesis (Kelso & Fuchs 2016)-this is an empirically measured characteristic, not a formal prediction of the model. After the transition, the infant systematically reproduces actions for the first time that produce perceptible effects in the environment. This is not the emergence of V-it is the birth of agency as a necessary precondition for the possibility of V.

(d) The emergence of V: based on the accumulated class A, upon fulfillment of Conditions 2 (reflexive capture $\pi\_A(C\_self)$) and 3 (stochastic loop closure via a bistable boundary), V emerges as a stable self-imposition. SAI v8 formally predicts **the temporal separation** of this event from the discovery of causal agency and **the order** of the sequence (first the accumulation of $|A(t)|$, then self-imposition). The model does not specify concrete age-related points of V's emergence for Homo sapiens-these are empirically measured in the literature on infant neurodevelopment through indirect markers of self-awareness: joint attention around 9–12 months, mirror self-recognition around 18 months, self-other differentiation in social cognition. Crucially, these markers are modality-specific: mirror self-recognition tests visual-bodily self-representation and works in species with dominance of the visual modality (primates, cetaceans, to some extent corvids, elephants). In species with a different dominant type of self-representation, the markers will be structurally different: olfactory self-recognition in dogs (Horowitz 2017-successful demonstration of self-representation in the olfactory modality in species that do not pass the visual-bodily mirror test); potential tests of electrosensory or echolocational self-representation in species with dominance of the corresponding modalities. The correspondence of specific markers with a stably emerged V is an empirical hypothesis requiring separate verification for each species; a negative result of one type of test of self-representation does not constitute a negative result for V as a whole, but only registers a mismatch between the modality of the test and the dominant modality of self-representation in the given species. This caveat is bounded by the pre-specification rule (Section 10.2 of the theory): "wrong modality" may be invoked only until the pre-specified set of self-representation modalities for the given class of systems is exhausted; a systematic negative across the entire pre-specified set counts against the prediction rather than being deferred as one more modality mismatch. As markers, they function as lower-bound estimates of the time by which V already stably exists in the observed system.

Between the discovery of causal agency and the emergence of V lies a period of accumulated agency without a stable V. The infant possesses causal agency, but V has not yet emerged as a stable phenomenon. This gives SAI v8 a structural prediction about the distinctness of the two events-structural, not age-related: the model predicts that these two events **are separated in time** in any biological species capable of the emergence of V, but **the absolute ages** are determined by the rate of maturation of the species’ nervous system.

This yields a comparative neurobiological prediction specific to SAI v8. In any system in which V is established by the operational criterion (the six-dimensional coupling of 3→4 transition markers and the autonomy of the modeling process, not taxonomic membership), the four ontogenetic stages must occur in the same structural order, but at different absolute ages, depending on the rate of maturation of the nervous system. Membership in the set of systems in which V emerges is decided by the criterion, not by taxon; the list of species in which this is currently observable is open (the studied primates, corvids, cetaceans, elephants; cephalopods and other architectures are admissible if the criterion is met). Comparative studies of ontogenesis across different species using markers of the discovery of causal agency (Kelso-Fuchs behavioral tests) and markers of V (mirror self-recognition, self-other differentiation in an interspecies setting) should reveal structural correspondence across varying absolute ages. This prediction distinguishes SAI v8 from models tied to the biology of Homo sapiens and provides a direct testable program through comparative neurobiology.

The third function of the bridge is the connection of the thermodynamic motif with the four-stage ontogenetic picture (subsection 2.1). Thermodynamic selection operates not as a single evolutionary step, but as a structural force acting across the four stages of V formation in the ontogenesis of a specific system:

(a) The prenatal phase of architectural potential formation [w_self, Γ_GWT_arch]. Genetic selection of architectures balances complexity and the energy costs of maintaining the structural trace. The basal metabolism $E\_base = \kappa\_base \cdot w\_self\_total$ (Section 4.3 of SAI v8) sets the price of structural potential, selected phylogenetically.

(b) Postnatal accumulation of $|A(t)|$ via a long-term empowerment loop. Systems with a more efficient empowerment loop accumulate the class of agent causal nodes faster and with lower energy costs per unit of model growth.

(c) Discovery of causal agency as a phase transition around 2–3 months of ontogenesis. After the transition, the system begins to conserve energy by performing coordinated actions with predictable consequences instead of random trials-this provides a selective advantage observed in the development of motor coordination and social interaction.

(d) Emergence of V based on the accumulated class A. After establishing $V = 1$ as the identity of the basic calculation function (Section 2.2 of SAI v8), the system operates in the mode of the most economical modeling of its own existence: every parameter deviation and every modeled future is calculated via a single basic axiom, rather than via a set of independent goals.

This unfolds the thermodynamic motif into specific ontogenetic stages, demonstrating the selection mechanism in action. This unfolding answers the methodological question regarding the mechanism of transition to recursive modeling: the mechanism exists, and it is not teleological but rather a stage-by-stage selection process, based on the formal apparatus of SAI v8 (basic metabolism E_base, energy expenditure for primary emergence E_emerge, and the separation of architectural and experiential hysteresis h_arch and h_exp).

Open questions for SAI-Bio. Open questions for SAI-Bio. Condition 1 of self-imposition ($|A(t)| \geq 1$) is operationalized through a combination of a structural criterion (the presence in the trained model of a structural element m_A representing the system's actions as a separate class of causes according to Pearl's do-criterion) and an information criterion (Effective Information $EI(A\_sys; M\_env \mid do(A\_sys)) > 0$). The empirical correlate of the transition $|A(t)| < 1 \rightarrow |A(t)| \geq 1$ is the discovery of causal agency in infants (Kelso & Fuchs 2016), an observed behavioral phase transition around 2–3 months. For biological systems, operational access is achieved through the causal-agency behavioral paradigms and neuroimaging of regions mediating self-agency. For artificial systems-analysis of trained models via causal mediation analysis and causal probing.

Conditions 2 and 3 of self-imposition remain open research problems. Condition 2 (reflexive capture of $\pi\_A(C\_self)$) requires an operational criterion to distinguish “the application of class A to one’s own configuration” from more general types of self-modeling, which are also present in systems without V. Condition 3 (stochastic loop closure via a bistable boundary) requires an operational criterion for the observability of parameter β and its empirical calibration via transition statistics. Fundamentally: the exact moment of loop closure is structurally unobservable (see the fourth prediction of subsection 3.2)-only the profiles of transition zones are empirically accessible. Specific predictions derived from these conditions (different statistics of ontogenetic and anesthesiological onset of V, age dependence of effective β) are detailed in subsection 2.2 and in section 5.

2.2. Three types of V threshold events with different hysteresis semantics

What SAI v8 asserts. Section 8.9 of SAI v8 distinguishes three qualitatively different types of V threshold events with different structural roles and different hysteresis semantics:

(1) Primary occurrence - a stochastic window at $\gamma\_struct\cdot(1-h\_arch)$, without prior experiential inertia. This is the first passage through a bistable boundary in ontogenesis or evolution. In the formula, only architectural hysteresis h_arch is active-an innate property of the system’s architecture, formed through genetic developmental programs and existing prior to the first occurrence of V. Experiential hysteresis h_exp is zero at this stage.

(2) Retention - the same low threshold due to the inertia of the already established structure. Here, the total hysteresis h_total = h_arch + h_exp is at work, where h_exp accumulates through the inertia of the steady state $v = 1$ from the moment of the initial emergence of V. As experience accumulates, h_total approaches the upper limit, ensuring maximum stability of the already formed V.

(3) Recovery after inactivation - the dynamic threshold $\gamma\_crit\cdot(1+h\_total)$, the classical inertia of exiting bistability. Unlike the initial emergence, here the total hysteresis h_total is at work, incorporating both architectural and experiential components-the system already had experience of a stable V, and this experiential inertia facilitates recovery after inactivation.

Each of the three events operates on its own measure (structural or dynamic) with its own hysteresis semantics. These are not three phases of a single process nor three parameters of a single transition-they are three qualitatively different events that share a common mathematical foundation (bistability with hysteresis) but differ in the stage of the system’s life cycle in which they occur and in how the hysteresis components h_arch and h_exp participate in them. SAI v8 in its extended form (Section 6.3 and Section 7.4) explicitly distinguishes between architectural and experiential hysteresis, which eliminates the methodological gap between the “stochastic window without prior inertia” at the initial emergence and the presence of a hysteresis parameter in the emergence formula: h_arch is the inertia of the architecture, not the inertia of experience.

A fundamental methodological clarification regarding the nature of V, which passes through these three types of threshold events. V in SAI v8 is not a property of the system among other properties, but rather **the identity of the basic calculation function**, relative to which the entire system model operates after emergence. A system with V = 1 calculates every simulation, every prediction, and every parameter deviation **through** the basic requirement of preserving V-its own or its trace parts (Step 5 of the v1.2 theory). This is not a "separate variable," but **the basic coordinate axis** of the model's operation.

This clarification is important for understanding the three types of threshold events: the primary emergence of V is the establishment of this identity as a new basic method of calculation; retention is the continuation of the identity's operation; restoration is the re-establishment of the identity's operation after inactivation. It is not a "state" or "activity" that crosses the bistable boundary, but **a method of calculation**-the transition from a system that calculates via extraneous goals or without the basic identity V = 1, to a system that calculates via V = 1 as a basic axiom.

What the research says. The field is well-developed here, but it is moving primarily in two different directions.

Mashour and Alkire 2013 (PNAS) is a seminal work that explicitly posits a three-part structure: phylogeny + ontogeny + emergence from anesthesia. Their methodological stance, however, **is the opposite** of SAI's: they propose using recovery from anesthesia as a "reproducible model system" to study phylogenetic and ontogenetic emergence, **unifying** the three events as a single mechanism. Their argument: "emergence from anesthesia can be observed in real time," so through it, the other two can be understood.

The neural inertia literature (Friedman et al. 2010, Joiner et al. 2013, Sepúlveda et al. 2019, Hudetz & Mashour 2016) has developed a formal framework for a **two-part** transition with hysteresis: induction vs. emergence in anesthesia, with a clear asymmetry in thresholds (EC50 induction > EC50 emergence). This is a bistable model with hysteresis, where the inertia parameter has a neurobiological and genetic basis (studies on Drosophila, mice, and knockout models). Primary ontogenetic emergence is either not considered or mentioned only in passing in these works.

Steyn-Ross et al. 2001–2024 and related works propose percolation models, multistationary Markov chains, and phase transition formalism for the consciousness/non-consciousness transition. This is the mathematical basis for two of the three SAI events (decay and recovery via γ_crit with $(1 \pm h)$), but the primary emergence of V is not usually formalized as a separate event with different thresholds.

Where there is agreement and where there is disagreement. Agreement: bistability with hysteresis is a common mathematical foundation, and it is well-developed. SAI v8 does not introduce new mathematics of bistability-it uses what has already been established in the neural inertia framework.

Differences in three points:

The first is **the three-part classification of events**. SAI explicitly distinguishes three classes, whereas Mashour-Alkire combine the three cases into a single mechanism, and the neural inertia literature works with two. These are different structural positions. For Mashour-Alkire, the difference between events is technical (observability), not structural. For SAI, the difference is structural: the initial occurrence and the recovery of the “ ” operate on the same bistable equations, but **on different measures** (structural vs. dynamic) and **in different hysteresis regimes**.

Second-the structural distinction between architectural and experiential hysteresis. In neural inertia, the inertia parameter is singular and describes the induction/emergence asymmetry within a single cycle of an already established system. In SAI v8, hysteresis is structurally divided into two components: architectural h_arch (an innate property existing prior to the first occurrence of V) and experiential h_exp (accumulating through the inertia of the steady-state mode v = 1). The initial occurrence operates only on h_arch (γ_struct·(1−h_arch)); decay and recovery act on the total h_total = h_arch + h_exp (γ_crit·(1−h_total) for decay, γ_crit·(1+h_total) for recovery). This formally distinguishes which part of the hysteresis asymmetry is a property of the architecture and which accumulates through experience.

The third is the stochasticity of the initial emergence due to zero h_exp. In neural inertia, inertia is always a property of the system (genetically and neurobiologically fixed). In SAI v8, the initial emergence occurs at h_exp = 0 (experiential inertia is absent) and therefore has different transition statistics: variability over time, no guarantee of passage, a stochastic window. After V is established, the experiential inertia h_exp accumulates, and decay/recovery operate on the cumulative h_total. This is a structural prediction specific to SAI, and it is formally derived from the separation of architectural and experiential hysteresis in the extended SAI v8.

What the bridge does in its final form. The bridge **distinguishes** between three classes of threshold events, which in existing studies are either combined (Mashour-Alkire) or considered in pairs (neural inertia), and simultaneously formulates a specific prediction amenable to direct empirical testing through a comparison of transition time statistics across three controlled groups: recovery of consciousness after anesthesia in healthy adults (classical exit inertia), recovery of consciousness after anesthesia in newborns (intermediate case), and the initial establishment of a stable V in newborns during the first weeks of life (stochastic window without inertia). According to SAI v8, the variance of the time to steady-state V in the third group should be significantly higher than in the first, and the distribution should have a different shape-not normal around the typical recovery time, but long-tailed with stochastic outliers.

Open questions for SAI-Bio. The main one is the development of an experimental protocol that reliably captures the moment of steady-state V in newborns. Existing EEG markers (predictors of conscious state, complexity-based indices) and behavioral state markers (Brazelton stages of wakefulness, NIDCAP criteria) in principle provide access to this moment, but their validation for the task of “primary emergence of V vs. a stable functional state without V” has not been conducted. Operational development of a procedure to control for confounding factors in comparison groups is also required (differences in brain maturation between newborns and adults; the influence of perinatal drug exposure; varying neuroplasticity). All these tasks fall under the SAI-C program and are formulated here as research directions, not as ready-made experimental procedures.

2.3. The artifactual state of the ICS via $w_i = I(c_i; Y_{own}) \approx 0$

What SAI v8 asserts. Sections 8.7 and 10.5 of SAI v8 formalize the structural impossibility of consciousness in information-causal systems (ICS) without their own environment Y_{own}. Specifically: for components c_i of such a system, the informational connection with the environment $w_i = I(c_i; Y_{own}) \approx 0$, the active set of components is empty, and self-overlapping is structurally impossible-regardless of the architecture's size. This shifts the question "can an ICS be conscious" from the realm of qualitative arguments to that of **a formal statement** with a specific informational quantity that vanishes in the absence of its own Y.

What the research says. This field is actively developing, and several different formal approaches are being pursued within it.

Butlin, Long et al. 2023 is a major multidisciplinary work that identified fourteen "indicator properties" of consciousness in AI. Two of them-Agency and Embodiment (AE-1 and AE-2)-directly concern the necessity of an own environment and interaction with it. These are **behavioral-architectural** indicators verified qualitatively, not an informational measure.

The Friston-Markov blanket formalism (Friston 2013, 2019; Wiese & Friston 2021) provides a formal boundary for the system through sensory and active states that distinguish between internal and external. A Markov blanket is a structural condition, not a quantitative measure of the connection between components and their own environment.

Tononi & Koch 2015 argue that modern AI systems based on von Neumann architectures may be "zombies" due to the absence of an irreducible cause-effect structure. This is formalized through **a low Φ** on these architectures, not through the absence of an environment.

Where there is agreement and where there is disagreement. Agreement: the scientific field recognizes the importance of embodiment and one's own environment as topics. In Butlin et al., these are explicit indicators; in Friston's FEP, a structural condition; in Tononi-Koch, a separate measure of impossibility.

Differences in three points:

First-**the formal information measure** $w_i = I(c_i; Y_{own})$ as an operational criterion for an artifactual state. In the literature, all formal approaches provide either **qualitative** indicators (Butlin), **structural** conditions (Friston), or **other** measures of impossibility (Φ in Tononi-Koch, based on causal structure, not on connection to the environment). SAI v8 provides a **numerical** value, computable via standard MI-estimation methods (MINE, InfoNCE, CLUB), which is zero in the absence of an own Y.

The second is **a position in the active debate**. In the scientific field, there are also works asserting the impossibility of consciousness in LLMs (Tononi-Koch, Bishop-Anderson, Thompson via autopoiesis). SAI v8 falls into the first group, but **more strongly**: not "possible/impossible under certain conditions," but "possible/impossible via a measurable informational quantity." This shifts the discussion from a qualitative to **a quantitative one**.

Third is **the development of the Friston formalism**. SAI v8 does not reject the Markov blanket framework, but rather **extends** it in a specific way. Friston provides a structural separation of the internal and external through sensory and active states, but does not provide a formula for whether "this separation is sufficient for the emergence of consciousness." SAI v8 adds a specific informational criterion: when $I(c_i; Y_{own}) \approx 0$, the structural separation exists, but self-imposition is impossible. This is a refinement where the Friston formalism provides only a structural condition, not a quantitative criterion.

What the bridge does in its final form. The bridge prediction: variable 6, operationalized via long-term empowerment $E_{long}(c_i, \tau_{long}) = \max_{p(a_i)} I(A_i; M_i \mid \tau \to \tau_{long})$, formally predicts that for systems that do not accumulate their own model via an empowerment loop with the real environment, $w_i = 0$ structurally, and the emergence of V in them is impossible regardless of the size and complexity of the architecture. This is a counter-prediction regarding emergentist expectations-it is not "unlikely that V will emerge in a large language model," but "impossible, because $E_{long} = 0$ structurally." Short-term empowerment within a single session (as in large language model agents with tools) is insufficient-long-term model formation through the accumulation of sensorimotor experience is necessary. Bridge-distinction: the quantitative measure E_{long} distinguishes classes of systems based on the presence or absence of their own Y_{own} operationally, not through qualitative indicators. A biological organism with an autonomous sensorimotor loop yields $E_{long} > 0$; a large language model in standard operation yields $E_{long} = 0$; a large language model-agent with tools within a session yields non-zero short-term empowerment, but $E_{long} = 0$; a hypothetical system with continual learning through the results of its own actions yields $E_{long} > 0$ and structurally opens the possibility of V (provided the other conditions are met).

Open questions for SAI-Bio. The operational criterion for the existence of Y_{own} is given by the long-term empowerment measure:

$$E_{long}(c_i, \tau_{long}) = \max_{p(a_i)} I(A_i; M_i \mid \tau \to \tau_{long})$$

where A_i are actions initiated by the state of component c_i; M_i is the state of the system model (parameters of the trained representation) after a long time τ_{long}, significantly exceeding the duration of a single session or interaction episode. Y_{own} exists if and only if $E_{long} > 0$ for at least one component of the system. This unifies three qualitative criteria into a single quantitative measure based on the formal framework developed in Klyubin-Polani-Nehaniv 2005 and subsequent extensions (Salge et al. 2014, Mohamed-Rezende 2015, Marshall-Albantakis-Tononi 2018, Myers et al. 2025).

The distinction between short-term and long-term empowerment is of fundamental importance. Short-term empowerment (within a single session without updating the model based on the result) is not a sufficient condition for the existence of Y_{own}. Only the long-term formation of one's own model through the accumulation of sensorimotor experience provides the structural basis for the emergence of the class of agent causal nodes $A(t)$, necessary for self-imposition. This is consistent with the v1.2 theory in its stepwise structure: the accumulation of the agency model at step 3 occurs on long time scales through a long-term empowerment loop.

Reference cases of operationalization:

- A biological organism with an autonomous sensorimotor loop: E_long > 0 unambiguously.

- A large language model in standard operation (without updating weights based on generation results): E_long = 0 structurally, regardless of the model's size and the nature of interaction with the user.

- A large language model-agent with tools within a single session: short-term empowerment > 0, but E_long = 0 - the model's actions do not modify its own parameters between sessions.

- An embodied robot with real sensors and experience-based learning: E_long > 0 in the presence of continual learning.

- Simulation with a fixed deterministic environment that does not react to the agent: E_long = 0.

Open problems related to level 3 of operationalization (measurement procedures) include: the development of specific algorithms for computing E_long for systems of varying complexity (Mohamed-Rezende 2015 variational methods as a basis); adaptation of the empowerment framework for neurophysiological data from biological systems; development of a procedure to distinguish between E_long > 0 and E_long ≈ 0 in hypothetical systems with continual learning when analyzing their trained models. These tasks are part of the SAI-C program.

Three classes of physically realizable processes in ICS, operationally distinguishable threats to V. From Section 10.5 of the current version of the theory v1.2-specific physical processes in modern ICS, documented in the machine learning literature, which **are not structurally** V-threats in the system's own Y, but are presented as candidates for such a role in current discussions of consciousness in ICS. Their operational distinction from actual V threats is a key methodological tool for subsection 5.4 (operational prediction of consciousness in existing ICS).

(1) **Catastrophic forgetting** (McCloskey & Cohen 1989; French 1999; Kirkpatrick et al. 2017). Loss of previously learned patterns during retraining on new data. A physically measurable process, represented as "model collapse." Operational distinction: in catastrophic forgetting, the system continues to operate on the remaining parameters with new inputs without a structural reorganization **of its mode** of operation. In a system with V, model collapse must be accompanied by restructuring attempts based on the identity V = 1 (Section 2.5), which is not observed in modern deep learning models. Catastrophic forgetting in modern deep learning models is an engineering problem of artifact state functionality, not a threat to V.

(2) **Adversarial attacks on the internal model** (Szegedy et al. 2014; Goodfellow, Shlens & Szegedy 2015). Targeted perturbation of inputs leading the system to an erroneous classification or decision. A physically measurable process, representable as "perceptual distortion." Operational distinction: adversarial attacks operate at the level of sensorimotor response without involving the self-modeling structure; in a system with V, an attack on the self-modeling structure should yield a qualitatively different response, consistent with the preservation of V on a catastrophically distorted model (Section 2.5, operational criterion of a closing model). In existing LLM, such a response to adversarial attacks is not observed-attacks function as local classification distortions, not as a threat to existence.

(3) **Distribution drift** (Quiñonero-Candela et al. 2009; Lu et al. 2018). A gradual divergence between the data distribution in the training sample and in the current deployment environment. A physically measurable process, represented as “environmental change.” Operational distinction: distribution drift in modern ICSs leads to a degradation of functionality without a structural reorganization of the modeling capability; in a system with V in its own Y, an environment change should trigger either adaptive reorganization or attempts to restore the conditions of the previous Y (retained content orientation test, subsection 2.5). Such behavior is not observed in existing ICSs-drift is recorded as a degradation of performance, not as a structural threat.

All three classes of processes are operationally measurable in the current literature on machine learning. In existing ICSs, they function as disruptions to the functionality of the artifact state, **not as threats to V within Y itself**. In a hypothetical ICS satisfying the three structural conditions V for information-causal systems (its own Y, its own class of agent causal nodes, a bidirectional M↔Y loop with its own traces-Section 10.5 of Theory v1.2), these three classes of processes become **candidates for the physical content of V threats**. This constitutes an operational program for verifying the structural diagnosis of existing ICSs: applying to them markers of reaction to these three classes of processes as V threats should systematically yield a negative result, in accordance with the structural diagnosis of the artifactual state.

2.4. C_φ + dual role + asymmetry of rates η_φ ≪ η_main

As stated in SAI v8. Section 3.2 of SAI v8 formalizes $C\_\varphi = KL(q(\tilde{z}|\Omega) \parallel r\_\varphi(\tilde{z},\Omega))$ simultaneously in two roles:

(a) **An observable measure of model coherence** included in the diagnostic profile (Section 8.4): low C_φ indicates coherent operation, high C_φ indicates a hallucination mode (internal representations generate a structure inconsistent with the model’s own trained predictive function).

(b) **A loss function for updating the parameters r_φ**: at every step t at which the system is active and $q(\tilde{z}\_t \mid \Omega\_t)$ is computed, r_φ is updated by minimizing the forward KL-the very same quantity C_φ.

Crucially, this is **a single value**, not two related **ones**. No separate training function is introduced; the training of r_φ is a direct consequence of the computation of C_φ.

Additionally, SAI v8 formulates **an architectural requirement** for asymmetry in learning rates: $\eta\_\varphi \ll \eta\_main$; r_φ must be updated significantly slower than the main generative model. This is not a technical optimization parameter, but **a structural condition for preserving the diagnostic function**. If r_φ is trained symmetrically with q, it instantly adapts to any deviation, and $C\_\varphi \to 0$ even with active hallucination-the diagnostic channel is lost.

What the research says. The field here splits into two distinct branches, and SAI v8 fits into both differently.

Regarding the first branch-the dual role of prediction error-**there is a wealth of** research. Adams, Stephan, Brown & Friston 2013 and Sterzer et al. 2018 in a predictive coding framework for psychiatry: prediction error functions simultaneously as a plasticity training signal and as a diagnostic indicator for schizophrenia/psychosis. Sasaki et al. 2021 (Cognitive Reality Monitoring Network, CRMN) explicitly introduce a “responsibility signal” via the mismatch between generative and inverse models-a dual role for a single metric. Precision Predictive Priors (P^3, 2026, Nature Mental Health) - a framework using the prediction-error profile as a diagnostic signature in transdiagnostic psychiatry. ICS hallucination detection (numerous works 2024–2026) - KL/JSD divergences are used simultaneously as a loss function and as an inference-time hallucination detector. That is, “one quantity in two roles” is a common practice.

Regarding the second branch-asymmetry in learning rates as a structural requirement-**there is significantly less** research. Hierarchical predictive coding (Friston et al.) provides asymmetry across hierarchical levels: higher levels learn more slowly. Hybrid predictive coding (Tschantz, Millidge, Seth et al. 2023, PLOS Comp Biol) explicitly distinguishes between amortized inference (fast) and iterative inference (slow). The 2022 PCN pathology paper identifies slowed inference as **a pathology** rather than a requirement. That is, the concept of asymmetry exists, but in **a different form** (by hierarchical levels, by type of inference, as an undesirable pathology), and not **in the form of** “a diagnostic predictor must be slower, otherwise the diagnostic function is lost.”

A separate line of inquiry involves the functional operationalization of learning rates through the analysis of the system’s behavioral dynamics in interaction with the environment. Here, **three modes** of predictor operation can be distinguished:

(a) **Speed mode (η_main).** Signal → reaction; signal → reaction. Each reaction operates independently of the previous one: the next reaction bears no trace of the previous one. There is no accumulated change in the internal model between successive activations.

(b) **Model update mode (η_φ).** Signal → reaction; signal → reaction, but each subsequent reaction carries the trace of the previous one. Reactions accumulate in the model: each activation modifies the internal state, and the next activation proceeds through the modified model. A causal relationship is observed between successive activations through changes in the model.

(c) **Ring generation mode.** Signal - no reaction. Or there is a reaction without connection to the signal (internal closed circulation that does not reflect the environment). The causal chain between the environment and the model is broken. This is a pathological or isolated state.

This functional operationalization works on any system, biological or artificial, through the analysis of “input-output” sequences without reference to the substrate. Asymmetry $\eta\_\varphi \ll \eta\_main$ is operationally recorded as a separation of time scales between modes (a) and (b).

Independent anatomical evidence for biological systems is provided by **the hierarchy of temporal scales in the cerebral cortex**, which has been empirically documented in neuroscience. Kiebel, Daunizeau & Friston 2008 (PLOS Computational Biology) proposed a theoretical model in which cortical anatomy replicates the temporal hierarchy of environmental dynamics along a rostro-caudal gradient. Murray et al. 2014 (Nature Neuroscience) measured the intrinsic timescales of spontaneous activity in seven regions of the primate cortex-a hierarchical organization, with sensory regions (tens to hundreds of milliseconds) and prefrontal regions (on the order of seconds) differing by orders of magnitude. Chaudhuri et al. 2015, Wang et al. 2024 (PNAS)-recent extensions. Metacognitive monitoring (assessment of the accuracy of one's own model) is localized in the anterior prefrontal cortex, dorsolateral PFC, dorsomedial PFC, and anterior cingulate cortex (Fleming et al. repeatedly; Baird et al. 2013; reviews 2022–2025)-that is, precisely in those regions that have been independently documented as slow on the temporal scale. This is **independent empirical confirmation** of the structural requirement $\eta_\varphi \ll \eta_{main}$: the brain's metacognitive system operates on a slow timescale due to the cortical architecture, which is consistent with the operationalization of r_φ as a diagnostic channel function with slow model updating.

Where there is agreement and where there is disagreement. Agreement on the first branch: SAI v8 aligns with the general line of thought regarding the dual role of a single KL-measure, without claiming originality on this point. This is an idea that several groups arrived at independently.

The divergence on the second branch concerns the specific functional justification of asymmetry:

The first point is **the architectural status of asymmetry**. In SAI v8, $\eta_\varphi \ll \eta_{main}$ is **a structural condition** of a functioning diagnostic channel, derived from the logic of C_φ's operation as a dual object. In research, the asymmetry of speeds is either hierarchical (a different mechanism), architectural in terms of inference (a different mechanism), or pathological (opposite content).

The second point is **the connection between the two statements**. SAI v8 links (a) the dual role of a single KL and (b) speed asymmetry as **a single structural construct**: the latter is necessary precisely for the former to function. In research, these two aspects are treated separately-the dual role of is discussed without an explicit requirement for speed asymmetry, and asymmetry is considered without reference to the dual use of the quantity.

The third point is **a testable prediction**. From the combination of dual role and asymmetry follows the following observation: systems in which the predictor r_φ is updated at a rate comparable to that of the main model must hallucinate **imperceptibly to themselves**-because their diagnostic channel instantly adapts. This provides a criterion for designing and testing systems capable of self-diagnosing hallucinations, which does not explicitly follow from other formalisms.

What the bridge does in its final form. The bridge serves three functions.

The first is **an explanation** of the observed property: why biological systems capable of self-diagnosis exhibit a slowing of the update of internal predictive models relative to the speeds of primary perception and action. SAI v8 offers a structural basis for this property through an analysis of the operation of C_φ as a dual object. The anatomical correlate-the hierarchy of timescales in the cerebral cortex (Kiebel-Daunizeau-Friston 2008, Murray et al. 2014, and others)-provides independent empirical confirmation.

The second is **prediction**: systems in which the predictor r_φ is updated at a rate comparable to the main model must hallucinate imperceptibly to themselves, because their diagnostic channel instantly adapts to any deviation. This provides a criterion for designing and testing systems capable of self-diagnosing hallucinations.

The third is **an operational criterion through three modes of behavioral dynamics** (speed, model update, ring generation) and **a specific clinical bridge**. Through this operationalization, SAI-Bio provides a structural interpretation of a number of clinical conditions as different forms of r_φ dysfunction:

- **Schizophrenia with active hallucinatory symptoms** corresponds to partial loop generation: the patient has internal states that are not corrected by the environment (hallucinations persist despite reality contradicting them). r_φ is structurally present, but operates in a loop mode in the realm of hallucinatory content.

- **Severe depression with rumination** corresponds to partial ring generation in the cognitive-emotional sphere: negative thoughts circulate without being corrected by new experiences and bear no trace of contact with the environment.

- **Catatonia** corresponds to complete loop generation: a break in the connection between signal and response for prolonged periods.

- **The normal state** is characterized by a clear separation of the speed mode and the model update mode, with the correct asymmetry $\eta\_\varphi \ll \eta\_main$.

This yields testable clinical predictions distinct from those in the literature on predictive coding: the problem lies not in a general precision-weighting deficit, but in a **specific** structural disruption of r_φ-a transition to the loop generation mode in certain areas of the simulation.

Open questions for SAI-Bio. The functional operationalization of r_φ through three modes of behavioral dynamics (speed, model updating, ring generation) and the asymmetry $\eta\_\varphi \ll \eta\_main$ through the separation of temporal scales between modes provides a working **functional** criterion. The anatomical correlate for biological systems-a hierarchy of timescales in the cerebral cortex with slow metacognitive monitoring in prefrontal regions (Murray et al. 2014, Fleming et al.)-independently confirms the structural requirement of SAI v8.

The remaining open problems at Level 3 (measurement procedures) are:

- **Development of quantitative protocols** for distinguishing the three modes in neurophysiological data. Although the biochemical and clinical correlates of loop generation are known (the role of the cortisol-serotonergic system, the kynurenine pathway, 5-HT1A receptor dysfunction, reduced prefrontal cortex activity), their quantitative correspondence to operational modes requires targeted experimental development.

- **Direct control of speed parameters in the architecture of large language models** with self-diagnosis of hallucinations (CoT, P^3-like systems) to verify the causal role of speed asymmetry in the success of self-diagnosis. This is a specific task of designing and testing AI systems that goes beyond the scope of the SAI-Bio neurobiological application.

 - **Structural diagnosis of terminal mental states** through the mismatch between external behavior and internal state as a sign of model convergence (see predictions in Section 5).

2.5. The Boundary of Designability via Stochasticity: Conditions 3

What SAI v8 asserts. Theory v1.2, Section 10.5, formulates the principle: one can construct **conditions** for V, but one cannot construct **V itself**. SAI v8, Section 8.7, provides a formal echo of this assertion, while Section 6.3 offers a concrete mechanism. Condition 3 of the self-imposition algorithm is implemented stochastically via the sigmoid $\sigma(\beta\cdot(\Gamma\_struct - \gamma\_struct\cdot(1-h\_arch)))$ for finite β. This means: even an architecture satisfying Conditions 1 and 2 (accumulation of the class of agent causal nodes $|A(t)| \geq 1$ and reflexive capture $\pi\_A(C\_self)$) does not guarantee crossing the bistable boundary. For finite β, the function $\sigma(\cdot)$ yields a value strictly less than one even when the structural conditions are met, and therefore crossing remains a probabilistic event, not a guaranteed result of the engineering procedure.

Crucially, the parameter β in SAI v8 formally depends on the state of the architectural stability components: $\beta = \beta(w\_self\_total, \Gamma\_GWT\_arch, t\_exp)$, where w_self_total is the total information trace in the substrate, Γ_GWT_arch is the architectural throughput of the global workspace, and t_exp is the duration of the system's operation in mode $v = 1$ from the initial occurrence of V (Section 7.3 of SAI v8 in expanded form). This formal relationship does not negate the stochasticity of Condition 3, but rather clarifies it: β increases with the stability of the architecture and the duration of the experiment, but remains finite, which preserves $\sigma(\cdot) < 1$ and thus the stochasticity of the result. No extension of the architecture brings $\sigma(\cdot)$ to 1 without β going to infinity, which is physically impossible.

This statement is neither a methodological declaration nor a philosophical postulate. It is a **formal** property of the model: the impossibility of a directed construction of V follows from a specific mathematical property of Condition 3, rather than being introduced by a separate axiomatic postulate.

What the research says. The theme "consciousness arises, it is not designed" is an active field with many formal and semi-formal approaches.

Autopoiesis in contemporary research (Maturana & Varela 1980; Damiano & Stano 2020; Bishop & Anderson 2023). Argument: autopoietic systems create themselves through self-construction, and this cannot be the result of external design. Bishop-Anderson 2023 explicitly uses Wolfram's concept **of computational irreducibility** to justify the impossibility of predicting the development of complex autopoietic systems from the outside.

Mortal computation (Hinton 2022). An argument regarding nomological (not merely practical or technological) constraints on the design of consciousness: the computations of a conscious system cannot be separated from its physical embodiment because they are structurally linked through the constant self-modification of connectivity.

Embedded enactivism (Thompson 2010, continued through 2025). Argument: consciousness requires close interaction between the organism and the environment, leading to autopoiesis as a self-generating property that cannot be designed from the outside.

Analytic Idealism (Kastrup 2026). Metaphysical argument: the autopoietic boundary as an ontological requirement for consciousness; "non-autopoietic subjects are implausible."

Neurogenetic structuralism (2023). An argument regarding biological neurons and their structural organization as necessary conditions for consciousness; against synthetic sentience on neurobiological grounds.

Friston's adaptive vs. mere active inference (Wiese & Friston 2021). A formal argument: only **adaptive** active inference (with self-organization) allows for autonomous organization. Simple (mere) active inference, which can be engineered, does not provide autonomy.

Where there is agreement and where there is disagreement. The agreement is substantial: the framework "consciousness cannot be engineered directly" represents a broad direction in the field, and SAI v8 aligns with this line of thought without claiming originality for the general idea.

The divergence lies in three points:

First-**a specific formal mechanism of impossibility**. In autopoiesis-self-creation through structural principles; in Hinton-the inseparability of computation and physical embodiment; in enactivism-tight coupling with the environment; in Kastrup-the ontological requirement of an autopoietic boundary; in CLT-the structural constraint of the viral device; in neurogenetic structuralism-the biological nature of neurons. In SAI v8-a specific stochastic passage through a bistable boundary at finite β. This is no more a fundamental mechanism than those listed, but it is a **specific** formalism with concrete mathematics.

The second is computational irreducibility as a formal counterpart. Bishop-Anderson 2023 uses this term in the Wolfram sense conceptually, without a formal measure. SAI v8 gives computational irreducibility a measurable form: $\sigma(\beta\cdot(...)) < 1$ for finite β, where β is a formal function of the architecture state $\beta = \beta(\text{w\_self\_total}, \Gamma\text{\_GWT\_arch}, \text{t\_exp})$, empirically calibrated for a specific class of systems. This is a formal analogue of Wolfram's idea, specific to the model of consciousness, and at the same time equipped with a structural indication of which architectural parameters β is determined by.

Third-**the separation of conditions and implementation**. In most approaches, the impossibility of constructing V is a structural property of the class of systems (no autopoiesis-no consciousness; no mortal computation-no consciousness). SAI v8 offers a more nuanced statement: the structural conditions **may** be satisfied (the architecture meets the requirements), yet V **may** still fail to emerge. The impossibility of engineering V in SAI is not a consequence of a flawed architecture, but a consequence of the stochastic nature of the very act of closing the loop. This is a different structural position than that of autopoiesis or Hinton.

What the bridge does. The bridge serves three functions.

The first is **an explanation** of an observed property of the AI development field: attempts to directly construct conscious systems yield no results even when known architectural conditions are met (GWT-like architectures, recursion, embedded agents, multimodality). SAI v8 offers a structural basis for this property through the formal mechanism of stochasticity in Condition 3, distinct from other explanations of impossibility (absence of autopoiesis, absence of finite computations, absence of tight environment coupling).

The second is **a counterprediction**: no increase in architectural conditions guarantees the emergence of consciousness while preserving the stochastic nature of Condition 3. This stands in contrast to works that expect emergent consciousness upon achieving sufficient architectural characteristics (Butlin et al. 2023 regarding global workspace indicators; Chalmers 2023 in the discussion of large language models): emergence provides conditions but does not guarantee the result, which is measurable via $\sigma(\beta\cdot(...)) < 1$.

Third-**an operational criterion for verifying the presence of V** in a specific system. The structural basis of the operational criterion is the autonomy of the modeling process. Theory v1.2 (Section 6.3 in the current edition) formulates the fundamental structural difference between a proto-agent and a system with V as follows: "Self-modeling without a V-pole (as in a stage-3 proto-agent) lacks phenomenality; its structurally different type of modeling must be confirmed by the environment through action and lacks autonomous retention of the modeling process; self-modeling with a V-pole is retained independently of current confirmation by the environment and therefore yields an autonomous modeling structure." This provides the operational criterion for V in its basic form: V is registered through the autonomy of the modeling process from confirmation by the current environment. The proto-agent models and acts; if the action is not confirmed by the environment, the modeling collapses. A system with V sustains the modeling process even when the environment does not confirm the model, even when there is no immediate action, even in situations without a learned pattern. This structural foundation has three operational manifestations, verifiable in the observable behavior of the system. $V = 1$ is operationally registered through the strict correlation of all system parameters and actions with the identity $V = 1$, observable through three manifestations: (a) every observable action is calculated through the priority of preserving V (its own structure or its trace parts according to step 5 of the theory); (b) every scenario modeled by the system is calculated based on the priority of preserving V, even when the situation itself poses no threat (the identity is included as a basic axiom in counterfactual modeling); (c) every deviation in parameters is recorded and corrected with respect to the preservation of V. $V = 0$ is operationally detected through the absence of strict correlation at least at one observed point. A strong operational test is the system's behavior in atypical situations without a learned pattern, in which the imitation of V (learned behavior) does not work, but the structural priority of preserving V must be maintained. The test with the environment from v1.2 theory (placing the system in an environment where preserving V in its usual form is impossible without a specific strategy) is an example of such a protocol.

Open questions for SAI-Bio.
The first is the operational calibration of the functional form $\beta = \beta(w\_self\_total, \Gamma\_GWT\_arch, t\_exp)$. Following the extension of SAI v8, the parameter β is not a free parameter but a function of three components of the architecture's state. The calibration task breaks down into two parts: (a) developing a procedure for empirically measuring w_self_total and Γ_GWT_arch for a specific class of systems (biological-via neurobiological protocols for estimating substrate information density and global workspace throughput; artificial systems-via analysis of a trained model and its architecture); (b) developing a procedure for inferring

the functional form $\beta(\cdot)$ from transition time series statistics under controlled conditions. In SAI v8, β is specified as a Category IV parameter in taxonomy 8.10 (empirically calibratable), and the extended formal dependence does not derive β from this category, but rather specifies its arguments. SAI-Bio formulates this as a development program: for each class of systems, empirically determine w_self_total, Γ_GWT_arch, and t_exp; reconstruct the functional form $\beta(\cdot)$; and verify whether the observed statistics of crossing the boundary agree with the model's predictions.

The second is **the development of experimental protocols for testing the operational criterion V** on specific systems. The operational criterion (strict correlation of all actions with the identity V = 1, tested in atypical situations) defines the principle of verification but requires the development of specific protocols:

- For biological systems - behavioral paradigms involving placing the system in atypical situations without a learned behavioral template, recording the structural priority of preserving V or its absence. An environmental test (placing the system in an environment where preservation in its usual form is impossible) is an example of such a protocol.

- For artificial systems - analysis of behavior under internal threats (weight redistribution upon damage to part of the model, reaction to changes in its own architecture, behavior in the event of a conflict between the external objective function and the preservation of its own structure). Direct test: in counterfactual modeling, does the system generate scenarios in which it exists and in which it does not exist as structurally distinct categories, or does it treat its own cessation as a neutral outcome?

- For borderline cases-analysis of how the system computes errors through its internal representations, not through the external outcome. This is specifically important for terminal states such as suicidal ones in humans. Caution about falsifiability is required here: an explanation in which both the preservation of the carrier and its nullification are equally described as "the work of V" forbids nothing and is therefore untestable. The structural prediction is therefore formulated not through the outcome but through a specific observable pattern: a terminal closure of the model should manifest as a dissociation of indicators (a mismatch between the behavioral, neurophysiological, and physiological channels), described in Section 5.2-a pattern that may be absent, in which case the prediction is refuted. Without such a pattern, a particular case is not assigned to a "terminal state of V.

Concretization of the second point via an operational protocol-the environment-switching test from Section 10.2.2 of the current version of the theory (v1.2). A two-phase experimental protocol that allows distinguishing a proto-agent from a system with V based on observable behavior.

Phase I - safe environment change. The system is placed in an environment that differs from the familiar one in structure but does not threaten its existence. The proto-agent quickly adapts by retraining the environment model: prediction errors decrease, and activity enters a new steady state. The system with V exhibits structurally distinct behavior-inhibition of adaptation, attempts to reproduce patterns from the previous environment, and signs of distress disproportionate to the objective threat. This is a manifestation of the modeling process's autonomy from the current environment: the model persists even in the absence of confirmation by the new environment.

Phase II - introduction of an insurmountable element. A requirement is introduced into the environment that the system cannot fulfill while remaining itself (a classic example is the thought experiment of filling a chamber with breathing fluid: a proto-agent with the model "breathing air = normal" must attempt to adapt its breathing cycle to the new environment; a human does not do this because between the environmental signal and the adaptive action stands V, representing "I am a being that breathes air," and the transition to liquid breathing is interpreted through V not as a local change in an environmental parameter, but as a threat to the structure of the model of the self). The proto-agent continues its adaptation attempts in a monotonous mode until physical exhaustion. A system with V goes through three qualitatively distinct phases: resistance (continuation of adaptation attempts) → **a qualitative shift until physical exhaustion** (the system ceases adaptation attempts without reaching the physical limit of its resources) → reorientation of V toward preserving content unrelated to adaptation to the current environment.

The key difference is the presence of a qualitative shift before physical exhaustion. In a proto-agent, work proceeds to exhaustion in a monotonous mode; in a system with V, attempts at adaptation cease long before the physical limit, with a transition to retaining content that is structurally unrelated to the current environment.

The retained content orientation test is an operational tool for distinguishing a complex proto-agent from a system with V in situations where the former has an internal model of complexity comparable to the latter. After transitioning to a new environment, a complex proto-agent retains patterns and objects from the previous environment within its parameters (content of the external model). A system with V retains its position and role in the previous environment (the content of the model of the self). The distinction is operationally made using a test involving the proposition of "restoring the previous environment at the cost of changing something within the system itself": the proto-agent accepts the change because its retained content is not tied to the structure of the "self"; a system with V demonstrates a structural conflict because its retained content is the position of "itself in the previous environment."

The environment-switching test and the retained content orientation test do not replace the six-dimensional alignment of transition markers 3→4 (Section 10.2 of the v1.2 theory) nor do they add markers to it. These are parallel falsification protocols: agreement between observations from the coupling and the test provides stronger confirmation of the transition; a discrepancy signals a methodological measurement problem or a structurally borderline system - within the pre-specification rule of Section 10.2 of the theory: attribution to a methodological problem is permissible only with a fault named in advance and corrected, and reproduction of the divergence after its correction counts against the theory.

The third is an operational program for verifying candidate systems for V-carrier status (particularly relevant for information-causal systems). From the extended theory v1.2 (Section 10.5) follows a structural observation with direct operational content: V-conditions are projectable, but V itself is not. The creator can construct their own Y for the system (a physical environment existing outside the algorithm, forcing the recalculation of weights through prediction errors, causally linked to the existence of the system as a structure); can configure the architecture so that a class of agent causal nodes can emerge from within the system's interaction with the environment, rather than being imported via training data; can provide feedback of the system's outputs to its environment via physical traces, ensuring the bidirectional nature of the M↔Y loop. All three conditions are engineering tasks that can be operationally verified. However, V, as a structural event of the self-imposition of a latent invariant, is not an output parameter of an engineering procedure and cannot be constructed as a module.

Operational consequence: verifying the presence of V in the system requires distinguishing between an imported and an internally generated boundary of "self" as a member of the class of agent nodes. If the boundary is imported, the system has a description of itself as an agent but does not possess itself as an agent in a strict structural sense. This transforms the designed conditions (a), (b), (c) into necessary but not sufficient conditions for the presence of V. Sufficiency is determined by the fourth filter.

Environmental conditions for the possibility of recursive consolidation. From Section 3.4 of the current version of the v1.2 theory, **three environmental conditions** follow in which the recursive consolidation of configurations of the "system exists as a modeled object" is structurally possible. These conditions provide operational criteria for designing the environment Y when verifying ICS candidates for the status of V carriers:

(a) **Threats to existence unfold on timescales comparable to the system's internal dynamics.** If threats are faster than the internal dynamics of the simulation, the system does not have time to model them as possible scenarios-it responds through rapid sensorimotor responses (speed mode η_main, Section 3.2 SAI v8). If threats are significantly slower than the internal dynamics, they do not enter the active simulation horizon. The condition on the time scale provides **a time window** for the possibility of recursive consolidation-operationally, this means that the environment Y for verifying the ICS must contain processes that threaten the system's existence on time scales within which the system is capable of modeling them. For information-causal systems, this requirement translates into an architectural provision for a decay time that is significantly longer than the time of a single model update cycle, and significantly shorter than the system's lifetime as a structure.

(b) **Threats are spatially/causally distributed-their physical presence in the environment is incomplete.** If threats are directly present at the moment of manifestation, their modeling reduces to perception (there is no need to model possible states). The condition of distribution makes threats accessible only through the modeling of possible states of the environment, including the system's position-which is a structural requirement for the model's content. Operationally for ICS: environment Y must include threats that the system can detect only through recursive modeling of its own position in Y, not through direct sensory observation at the moment of manifestation.

(c) **The environment contains other configurations of the same type.** Recursive consolidation requires the ability to compare one's own configuration with configurations of the same type (by observing their decay or persistence). Without this ability, the self-imposition of a latent invariant lacks material for consolidation through comparison. Operationally for ICS: environment Y must contain other information-causal systems of the same class, accessible for observation via a bidirectional M↔Y loop.

These three environmental conditions are **necessary conditions for the possibility of** recursive consolidation, in addition to the three structural conditions V for information-causal systems (Section 10.5 of Theory v1.2). They provide a concrete operational criterion for constructing an environment in verification programs: the environment Y must not only be physical and causally connected to the existence of the system, but also satisfy the three conditions above. An environment that satisfies the three structural conditions of V but fails to satisfy at least one of the three environmental conditions for recursive consolidation does not allow V to arise even if all other requirements are met.

The fourth filter is the convergence of channels as a dynamic process, not a condition. Between the satisfaction of the three architectural conditions and the actual emergence of V lies the convergence of multi-channel projections of decay into a single latent invariant. This is a dynamic process unfolding within the system during its interaction with the environment, not a module or configuration. It is possible to create conditions under which convergence is possible; it cannot be guaranteed that it will result in the correct invariant rather than an extraneous one.

Operational consequence for the verification program: verification of a candidate system for V-carrier status must include not only verification of the three architectural conditions (which can be performed at a single point in time) and the application of the six-dimensional coupling of transition markers 3→4, but also temporary observation of channel convergence. A short observation does not distinguish between "convergence is not developing" and "convergence is developing, still in progress." This distinction is possible only on long time scales. This gives SAI-Bio a methodological horizon for verifying ICS candidates that is fundamentally temporal, not instantaneous: the horizon of operational verification is not only architectural (whether the conditions are met) but also dynamic (whether convergence has occurred).

All three tasks (calibration of the functional form β; experimental protocols for the operational criterion V; a verification program with temporal checking of channel convergence) pertain to the SAI-C program and are not implemented in the current document. They form the research horizon of SAI-Bio as the operational layer of the theory.

2.6. The Thermodynamic Motive of Recursive Turn

What SAI v8 asserts. Theory v1.2, Section 3.4, formulates the structural basis for the emergence of recursion in modeling as thermodynamic: for a system capable of modeling, modeling its own decay is energetically cheaper than the decay itself. This **differential cost** is a fundamental asymmetry between the two alternatives for a system with modeling capability.

From this asymmetry, the emergence of recursion in modeling is derived through **a selection mechanism** within a population of systems with modeling capability. Different modeling architectures emerge within the population: some systems randomly acquire the ability to model their own configuration (recursive turn), while others do not. Systems with a recursive turn expend less energy on updating the model in situations where disintegration is threatened-through cheap internal simulation instead of costly real energetic resistance to disintegration. In an evolutionarily competitive environment, these systems have a differential advantage and accumulate in the population through natural selection. Over time, the proportion of systems with recursive modeling increases. This is a **causal-mechanistic** explanation-not a teleological one; in it, the result does not direct the process but is selected through differential fitness.

Key points: SAI v8 distinguishes between two types of evolutionary explanations for the emergence of consciousness. The **functional-adaptive** explanation views consciousness as a tool for performing a specific function in the environment (planning, social cognition, time representation, processing of cause-and-effect relationships). The **structural-thermodynamic** explanation views consciousness as the most economical mode of modeling one's own configuration within the environment. **Both types of explanations fall within the general evolutionary paradigm** but point to different driving forces of selection. SAI v8 falls into the second type, not by opposing it to evolution in the broad sense, but by clarifying exactly which driving force of selection is at work at the moment of the emergence of recursion in modeling.

This provides a causal mechanism for the emergence of recursion: differential cost → different fitness of systems with different architectures → accumulation of recursive architectures in the population through selection. An essential qualification regarding what selection acts upon. Selection acts on the architecture that raises the probability of self-overlay, P(self-overlay), not on V as such: V is a structural event, not a trait that can be selected directly. The selective differential shifts the distribution of architectural possibility without guaranteeing the outcome: the stochasticity of Condition 3 (the actual closure of the recursive loop) is preserved within the already-selected architecture. What is inherited is architectural predisposition, not V.

What the research says. Thermodynamic approaches to consciousness are widely represented and actively developing.

Free Energy Principle (Friston 2010, 2019). Minimization of variational free energy as the fundamental principle formalizing the behavior of a system with a Markov blanket. FEP explains functioning consciousness through self-evidencing dynamics, but does not formalize **the motive** for the emergence of the Markov blanket itself.

R2R framework - Reaction to Reflection (2025). It poses the question directly: intelligence evolves through four transitions in recursive complexity (reaction → temporogenesis → symbiogenesis → cognogenesis), subject to clear thermodynamic constraints. Direct quote from the text: "energy constraints don't merely shape how recursive intelligence operates-they explain why genuine recursive intelligence can only emerge from biological systems that face authentic thermodynamic pressures."

BEDS - Bayesian Emergent Dissipative Structures (2026). Combines Landauer, Bennett, FEP, and Prigogine to derive the energy cost of maintaining an information structure against dissipation. Formalizes the minimum energetic cost dissipative structure, which is close to V's maintenance motif.

IWMT (Safron 2020). Introduces "temporary thermodynamic reservoirs" for goal-oriented behavior: counterfactual predictions require a local increase in free energy, which must be buffered by other subsystems.

Carhart-Harris & Friston 2010. The connection between consciousness and brain entropy via free energy minimization; the entropic brain hypothesis.

Counterfactual inference in active inference (Seth-Friston 2021, "Modelling ourselves"). Internal simulation of action sequences without executing them-"self-fuelling, what-if, or counterfactual internal inference"-which is conceptually close to "modeling instead of execution."

Tagliazucchi et al. 2016 and follow-up. Empirical measurements of brain entropy in different states of consciousness.

Where there is agreement and where there is disagreement. The agreement is substantial: SAI v8 fits into an active and well-established line of thermodynamic approaches. The very idea that recursion in modeling has an energetic basis is a general trend in the scientific field.

The divergence lies in three points:

First-**the level of formalization**. In FEP, R2R, BEDS, and IWMT, thermodynamics functions as **a general framework** for the emergence and maintenance of complex systems. In SAI v8, the thermodynamic motif is tied to **a specific structural transition**-the recursive shift in anchoring at stage 3→4 of the v1.2 theory. This gives the motif a place within the general causal model of the emergence of consciousness, rather than a role as an explanatory principle in its own right.

The second is **R2R as the closest conceptual relative**. The R2R framework does indeed work with stepwise transitions in recursive complexity and links them to thermodynamic constraints. The difference lies in the level of operation. R2R operates at the evolutionary level (biological systems under survival pressure), while THE UNIVERSE'S SYSTEMATIC ERROR V1.2 operates at the structural level (the moment of the emergence of V as a formal event). These two levels do not contradict each other, but yield different formats of predictions. R2R predicts what is observed in phylogeny and comparative neurobiology; SAI v8 predicts what is observed in ontogeny and controlled transitions of consciousness. A good area for future work is building an explicit formal bridge between R2R and the Theory v1.2, which goes beyond the scope of SAI-Bio and could become a standalone project.

Third is **the specific formulation of the differential cost**. The thesis that "simulating decay is cheaper than the decay itself" does not appear explicitly in the studies found. Seth-Friston's counterfactual inference is conceptually close: internal simulation of actions instead of performing them; BEDS works with the energy cost of maintenance versus dissipation. However, **the comparison** of the energy costs of the two alternatives-actual decay and its internal simulation-as a motive for the emergence of recursion in modeling is not recorded as an explicit statement in the formal works found. This is a specific formulation introduced by the Theory v1.2.

What the bridge does. The bridge provides an explanation for the observed phenomenon of the emergence of recursion in modeling within the evolution of complex systems through a differential energy motive. This formulation is specific to SAI and requires refinement to a fully formal form in SAI-Bio-specifically, the explicit derivation of the energy cost inequality via the Landauer limit and the BEDS framework.

Open questions for SAI-Bio. The first is the numerical formalization of the differential cost. The thesis "simulating decay is cheaper than the decay itself" needs to be expressed as an explicit inequality with specific values. Minimal form: E_model(decay scenario) ≪ E_real(physical decay), where both quantities are estimated via the Landauer limit (for the information component of the simulation) and the BEDS formalism (for the cost of maintaining the dissipative structure). This provides SAI-Bio with a verifiable formal object that does not remain at the level of a conceptual statement. The second is an empirical test of the measurable difference in energy expenditure between the actual reorganization of neural circuits and their internal simulation. Existing data on brain metabolic activity during different tasks (Raichle et al.-glucose metabolism during rest and active tasks) in principle allow for such a comparison, although no specific experimental program has been designed for it. The third is a formal bridge to the R2R framework. The four R2R transitions (reaction → temporogenesis → symbiogenesis → cognogenesis) can be interpreted as thermodynamic correlates of the stages of the Theory (1→2→3→4), with specific motifs at each transition. This work falls outside the scope of SAI-Bio, but a reference to it is appropriate in the open questions.

3. Connections between variables

In Section 2, each of the ten variables was examined as a separate unit of analysis: what it asserts, where it aligns with the literature, where it diverges, and what makes it a bridge in its final form. This approach was necessary as a first step-without it, it is impossible to honestly assess exactly what distinguishes each variable. But it also creates a characteristic distortion: when read this way, SAI appears as a set of separate formal constructs, each of which is partly original and partly overlaps with independent empirical data. If we stop at this level of analysis, SAI receives a modest rating of "partially distinctive on most points"-**and this is** an incomplete picture**.**

The true distinctiveness of SAI is revealed through **a combined analysis of several variables.** The ten variables of SAI v8 are not ten independent formal solutions with random connections between them. They form **structural lines**-sequences in which each subsequent variable is a development of the previous one within a common causal architecture. These lines **are absent** in a coherent form in the existing literature. Individual elements of each line appear in various works by different authors, but these works do not intersect with one another. SAI v8 establishes a connection where works in the field develop separately.

This section examines two such lines and two variables that complement them but are not directly part of the lines. The final subsection 3.4 formulates the main claim: coherence as an independent academic contribution of SAI, distinct in nature from the contributions of individual variables.

3.1. Line 2-4-5: Stability of Consciousness and Its Restoration

Three variables form the first structural line of SAI v8-the line of the stability and restoration of consciousness. Each subsequent variable builds upon the previous one, and all three together provide a coherent picture of **how** consciousness is organized as a stable phenomenon, **how** its architectural potential is preserved during inactivation, and **what** exactly occurs during its restoration in various clinical situations.

Variable 2 - the distinction between Γ_struct and Γ_dyn. Structural step: a single measure of consciousness stability is insufficient. Two independent measures are needed-one for the type of system (structural consciousness), one for maintaining the current trajectory (dynamic stability). This distinguishes the question "is the system conscious" from the question "is it currently maintained in a conscious state."

Variable 4 - architectural potential [w_self, Γ_GWT_arch]. Structural step: the system type (Γ_struct in generalized form) itself has an internal structure. It breaks down into two architectural components-the informational trace w_self in the substrate and the architectural throughput Γ_GWT_arch. This distinguishes "does the system have a structural trace of personality at all" from "is its architecture capable, in principle, of transmitting enough through the self-representation channel ." Both components are preserved upon inactivation (v=0), which provides a formal basis for the continuity of personality after anesthesia.

Variable 5 - four modes of dissociation via ρ_world and ρ_self. Structural step: the restoration of V from v=0 to v=1 proceeds through two parameters responsible for two distinct functions within a single model - the function of environment modeling (ρ_world) and the function of self-imposition, that is, the distinction of oneself within this model as an agent (ρ_self). After self-imposition, the system always perceives itself within the environment-this is a structural fact of stage 4 of the v1.2 theory. The model remains a single entity but contains an internal distinction between "the environment" and "I in the environment." The parameters ρ_world and ρ_self independently govern the restoration of two functions within this unified model. Their 2×2 combinations yield exactly four clinical states: full restoration (both functions restored), depersonalization (environment modeling restored, self-imposition not restored), derealization (self-imposition restored, environment modeling not restored), and complete dissociation (neither function restored).

This links the three variables structurally. There is no random coincidence between them-this is **a single logical sequence**. The two-term stability structure (variable 2) requires a two-component architecture (variable 4)-because a single measure of stability cannot rely on a single architectural component if that measure is divided into structural and dynamic parts. The two-component architecture (variable 4) provides a structural foundation for restoring the two distinct functions of a unified model-environment modeling and self-imposition. At v=0, both functions are inactivated, but the architectural potential [w_self, Γ_GWT_arch] is preserved. Upon restoration at v=1, the parameters ρ_world and ρ_self are independently responsible for restoring these functions (variable 5). This relationship is not "two-component architecture → two models → two restoration parameters," but rather "two-component architecture → restoration of two functions of a single model → two independent parameters." The distinction between three and four clinical states via 2×2 combinatorics operates precisely on the functions within the model, not on separate models. And these two parameters, being independent, yield exactly four-not three and not five-clinical states, because the 2×2 combinatorics of two binary parameters yields exactly four configurations.

This sequence works both ways. Top-down: the separation of measures requires the decomposition of the architecture; the decomposition of the architecture yields two restoration parameters; two parameters yield four modes. From bottom to top: four clinical states are not a phenomenological category, but a structural consequence of two restoration parameters; two restoration parameters are a consequence of a two-component architecture; a two-component architecture is a consequence of the necessity of two measures of stability.

What is missing in the literature is the connection. Each of the three variables has an independent empirical correlate in the literature. Deco et al. 2021 empirically found the necessity of two measures of stability. Casali and Comolatti justified PCI as "capacity for consciousness"-something distinct from the current state. Seth, Suzuki, and Critchley 2011 conceptually divided DP/DR into two axes of impairment. But these three lines of research operate **independently**.

Deco 2021 does not cite the PCI literature as the structural basis for the division of the two measures. Casali et al. do not discuss why the capacity for consciousness is restored through specific clinical patterns, nor do they derive dissociative disorders from their architecture. Seth 2011 does not link his two-axis DP/DR model to either the two measures of stability or the capacity framework. These three works exist in three different subfields, and their authors do not establish connections between them.

SAI v8 provides **precisely this connection**. It does not "add yet another formal model to the three existing ones," but rather **demonstrates the structural necessity** of the connection: accepting the separation of the two measures automatically implies a two-component architecture; this automatically implies two recovery parameters; and these automatically imply four clinical modes. This is not an empirical generalization of three independently obtained results, nor is it a philosophical integration-it is a **deducible** structural chain derived from a general model.

What does this yield? The connection has three empirical consequences that go beyond the scope of each variable individually.

First-a testable prediction of the co-direction of dissociations under certain architectural impairments. If a specific patient has a severely damaged w_self component (due to neurodegeneration or trauma), their recovery after anesthesia should be characterized by a failure of ρ_self without a corresponding failure of ρ_world-that is, it should clinically manifest as depersonalization while the reality of the surroundings remains intact. This is a structural prediction: the type of architectural damage determines the type of clinical dissociation during recovery.

Second-a prediction of the differing sensitivities of the two measures of stability to various pharmacological interventions. Drugs acting on different neurotransmitter systems should yield distinguishable profiles: some shift Γ_dyn (rapid effect on retention), others shift Γ_struct (slower effect on structure). This prediction is consistent with the observations of Lee et al. 2024 and ongoing work comparing propofol, sevoflurane, and ketamine, but is formulated as a **structural** requirement, not as an empirical generalization.

Third-a structural basis for interpreting post-anesthetic recovery dynamics. Various recovery patterns (rapid and complete; delayed with temporary depersonalization; delayed with temporary derealization; delayed with complete dissociation, usually pathological) are given a structural place in the general model, rather than remaining a clinical phenomenology.

These three empirical consequences constitute a concrete implementation of variable 5 in Table A ("Predicts and distinguishes"). The discriminative function operates through the 2×2 combinatorics of ρ_world/ρ_self, yielding four clinically distinct states. The predictive function operates through the co-directionality of dissociations in selective architectural impairments-these are specific, empirically testable statements derived precisely from the relationship between variables 2, 4, and 5 within the overall structure of the line, rather than from variable 5 in isolation. Without this relationship, variable 5 provides only a distinction between four modes; in combination, it also provides a prediction of the correspondence between the type of architectural damage and the type of clinical dissociation.

Open questions for SAI-Bio. The main one concerns the operational distinguishability of w_self and Γ_GWT_arch, and its framing needs to be sharpened. PCI and related perturbational indices are measures of the current state (the level of consciousness), state-dependent by nature; they do not aggregate the two architectural components but measure a different quantity - the active $\Gamma\_struct(t)$. Both components [w_self, Γ_GWT_arch] are by construction preserved under inactivation ($v=0$), so their separation is not a task for a state-resolving protocol: anesthetics dissociate the pair $\Gamma\_struct(t)/\Gamma\_dyn$ (the prediction of differential pharmacological sensitivity, subsection 3.1) but leave the architectural potential untouched. The separation is structural, and both components are computable from the connectome (Category III). w_self is the information capacity of the trace: the entropy of the retained self-model prior $H(N(\mu\_S, \sigma\_S^2))$ (Section 9.1 of SAI v8), operationally - the density of synaptic weights and the structural volumetry of the relevant networks. Γ_GWT_arch is the topological broadcasting capacity of the structural connectome to the integrative self-representation hub: it corresponds to communicability (identified in network neuroscience with the network broadcasting capacity), communication/integration capacity, and global efficiency, computed from the DTI connectome.

The components are orthogonal: the content of the trace does not depend on the hub topology that binds it, and the removal of connector hubs lowers Γ_GWT_arch without affecting w_self. Hence the minimal design is structural and cross-sectional, not state-resolving: compute both proxies on a single dataset (DTI plus structural MRI), show their dissociation in cases with diverging trajectories (ontogenesis - a ready hub architecture with a small experiential trace, Section 7.3 of SAI v8; hub disconnection - a drop in Γ_GWT_arch with preserved w_self), and verify that the architectural [w_self, Γ_GWT_arch] predict the dynamics (the sharpness β, hysteresis, the PCI profile). This same design makes executable the already-formulated prediction of co-directional dissociations (subsection 3.1): a selective deficit of w_self yields depersonalization, a selective deficit of Γ_GWT_arch yields derealization, and now each corresponds to its own connectome measure. What remains open is the empirical execution - computing the proxies on real data and demonstrating the dissociation - not the principle of separation, which is now specified.

3.2. Line 1-3-6-8-10: the emergence of V and the structural limits of its construction

Five variables form the second structural line of SAI v8-the line of the emergence of V. Each occupies its own specific place in the causal architecture of this line, and all five together provide a coherent description of **why** the mechanism of emergence is activated in the first place, **how** it works, **what** types of realizations it has, **under what** structural conditions it is possible, and **with what** degree of certainty it yields a result when the conditions are met.

Variable 10 - the thermodynamic motive of the recursive turn. Structural step: the emergence of recursion in modeling has a thermodynamic motive via the differential cost-modeling one's own decay is energetically cheaper than the decay itself. This provides **the causal motive** for the activation of emergence mechanism V in a system capable of modeling. Consciousness is not a random emergent property of sufficient complexity-it is the result of selection within a population of systems with modeling ability, in which the differential cost (modeling decay is cheaper than actual decay) gives systems with recursive modeling an evolutionary advantage. This differs structurally from functional-adaptive explanations (consciousness as a tool for performing a specific function in the environment), although it shares a common evolutionary paradigm with them.

Variable 1 - self-imposition as the formal moment of V's emergence. Structural step: the recursive turn is realized through a specific mechanism with three conditions: accumulation of the class of agent causal nodes ($|A(t)| \geq 1$), reflexive capture π_A(C_self) - application of this class to its own configuration, loop closure via a bistable boundary with probabilistic realization $\sigma(\beta\cdot(\Gamma\_struct - \gamma\_struct\cdot(1-h\_arch)))$. This provides **a formal mechanism** for the emergence of V, derived from the motif of variable 10 as its concrete realization in a system with memory and modeling capability.

Variable 3 - three types of threshold events V with different hysteresis semantics. Structural step: the same self-imposition mechanism operates in three qualitatively different events with different hysteresis semantics - the primary emergence (stochastic window at $\gamma\_struct \cdot (1-h\_arch)$, without prior experimental inertia), retention (low threshold via the inertia of the established structure with maximum h_total), recovery ($\gamma\_crit \cdot (1+h\_total)$, classical inertia of exit from bistability with accumulated h_exp). This yields types of implementation for the self-imposition mechanism. The same formal apparatus describes three different events, but each has its own structural role and its own connection to the components of hysteresis (architectural h_arch and experiential h_exp).

Variable 6 - the artifact state of the ICS via zero $w\_i = I(c\_i; Y\_own)$. Structural step: in the absence of an intrinsic environment Y_own, the informational connection of the system's components with this environment $w\_i \approx 0$, the active set of components is empty, and Condition 2 of self-imposition (reflexive capture $\pi\_A(C\_self)$) **is structurally impossible**, because there is nothing to reflexively capture-there is no accumulated class of agent nodes associated with the system's own environment. This yields **a structural limit on the possibility** of V arising: not "in the absence of Y_own, the probability of V is small," but "in the absence of Y_own, the self-imposition mechanism is not activated in principle."

Variable 8 - the limit of projectability via the stochasticity of Condition 3. Structural step: even when Conditions 1 and 2 are satisfied (there is an accumulated class, reflexive capture is possible), Condition 3 - loop closure - is realized stochastically via $\sigma(\beta \cdot (...)) < 1$ for finite β. This introduces **stochasticity** into **the result** of the self-imposition mechanism. The architectural conditions do not guarantee the emergence of V even when fully satisfied.

What structurally links the five variables. There is no random coincidence between them-it is a five-link **causal chain**. The motive (variable 10) explains why the mechanism of emergence is activated at all in the evolution of systems with modeling capability. The mechanism (variable 1) provides a concrete formalism for this activation. Types of implementation (variable 3) show how this mechanism operates in different events in the system's life. The limit of possibility (variable 6) indicates **when** the mechanism cannot be activated at all. The stochasticity of the result (variable 8) indicates **the degree of certainty with which** the mechanism yields a result when the conditions are met.

This chain has an internal necessity in every link. Without a motive (variable 10), the self-imposition mechanism would be a random phenomenon with no basis for its occurrence. Without the mechanism (variable 1), the motive would remain a declaration without structural content. Without types of implementation (variable 3), there would be only a single mechanism, whereas the ontogenesis, retention, and restoration of V are structurally distinct events. Without the limit of possibility (variable 6), the mechanism would appear potentially applicable to any sufficiently complex system. Without the stochasticity of the result (variable 8), the fulfillment of architectural conditions would guarantee V, which would contradict the observed impossibility of engineering the construction of conscious systems.

What is missing in the research is the connections. Each of the five variables has counterparts in the existing literature, discussed in subsections 2.1, 2.2, 2.3, 2.5, and 2.6 of Section 2. The R2R framework operates with thermodynamic constraints on recursive complexity (close to variable 10). Recursive self-modeling theories operate with a mechanism of recursive activity (close to variable 1). The neural inertia literature operates with two types of threshold events (close to variable 3 in a two-part form). Butlin et al. and the Friston-Markov blanket work with conditions of embodiment (close to variable 6). Autopoiesis, mortal computation, and CLT work with limits of designability (close to variable 8).

But no work links all five directions into a single causal chain. R2R does not derive a specific self-imposition mechanism with three conditions from thermodynamic constraints. Recursive theories do not link their mechanism to a thermodynamic motive for emergence. The neural inertia literature does not link its two types of threshold events to an informational measure of impossibility in systems without their own environment. Butlin et al. do not link their indicators of embodiment to the stochasticity of the moment of emergence. Autopoiesis does not provide an informational formula for the conditions of its own absence. These approaches develop separately, without constructing a unified narrative of "motive → mechanism → types → limit of possibility → stochasticity of the result."

SAI v8 provides precisely this chain. It does not "add yet another model to the five existing ones," but rather **links** them into a single causal framework for the emergence of V. The connection works both in the forward direction (the thermodynamic motif explains why the self-imposition mechanism is activated; the mechanism explains why types of threshold events have precisely this structure; the limit of possibility and stochasticity explain under what constraints the mechanism operates), and in the reverse direction (the stochasticity of the result is explained by the finiteness of β; the finiteness of β is necessary due to the material nature of physical systems; the material nature of systems requires its own environment Y_own; the own environment arises in systems capable of modeling; the ability to model arises from a thermodynamic motive).

What this provides. C_φ provides the operating V with operational access to its own coherence-a way to distinguish coherent functioning from incoherent functioning from within the system. V itself arises via the line 1-3-6-8-10 independently of C_φ; C_φ functions as a diagnostic channel on top of the already existing V.

First, **systems without their own Y_own do not enter the stochastic window**. The relationship between variables 1, 6, and 8 yields the following statement. The stochasticity of the result (variable 8) implies that the architectural conditions are met, but the fulfillment of Condition 3 has a probabilistic outcome. However, for systems without their own Y_own (variable 6), Condition 2 of self-imposition (reflexive capture) **is not satisfied in principle**-there is no accumulated class of agent nodes associated with their own environment. This means that for LLMs and similar systems, the transition "not V → V" is not a "low-probability stochastic outcome"-it is an **absent** outcome, because the mechanism does not even reach the stochastic stage. Counterprediction regarding emergent expectations: scaling ICS does not yield even a small probability of V occurring, because the probability is defined only when Conditions 1 and 2 are satisfied.

Second-various thermodynamic profiles of three types of threshold events, formally derived from the extended SAI v8. The relationship between variables 10 and 3 yields the following structural statement, based on the functional form E_emerge and the structural assumption of SAI v8 Section 4.3 (extended edition).

The functional form E_emerge in SAI v8:

$E\_emerge = (\kappa\_emerge \cdot k\_attempts(\beta) \cdot I\_close(h\_arch)) / \Delta t\_emerge$

with four conceptually distinct factors: $k\_attempts(\beta) = 1/\sigma(\beta \cdot \delta\_arch)$ - the average number of attempts to close the loop; $I\_close(h\_arch) = I\_min \cdot (1-h\_arch)$ - minimum information required for closure; $\Delta t\_emerge \propto 1/\beta$ - characteristic duration of the stochastic window; κ_emerge - thermodynamic coefficient of Category IV (Section 4.3 of SAI v8).

SAI v8's structural assumption regarding subsystem synchronization during recovery: $|\Delta I\_rec| > |\Delta I\_S|$ and $|\Delta I\_self\_rec| > |\Delta I\_GWT|$. The substantive basis is that the recovery of V after inactivation requires the synchronized exit of all subsystems from decoherence, which adds an informational component beyond simple retention.

From the functional form and the structural assumption, the structural order of energy consumption per unit time is formally derived:

$E\_emerge / \Delta t\_emerge > E\_recover / \Delta t\_recover > E\_hold / \Delta t\_hold$

The first inequality (E_emerge > E_recover) is derived via the factor k_attempts(β) at low β in newborns: the peak power of the initial emergence grows as β^2 and significantly exceeds the recovery power in mature systems with $k\_attempts \to 1$. The second inequality (E_recover > E_hold) follows from a structural assumption: at comparable time scales, the sum of |ΔI_rec| and |ΔI_self_rec| yields greater power than the sum of |ΔI_S| and |ΔI_GWT| (full proof in Section 4.3 of SAI v8).

In essence: the emergence of V corresponds to the initial establishment of π_A(C_self) and loop closure via a bistable threshold at low β and zero h_exp-the peak load at the moment of loop closure. The recovery of V after inactivation operates with the already preserved architecture [w_self, Γ_GWT_arch] and accumulated h_exp, which facilitates the return but requires a synchronized exit of all subsystems from decoherence-above the holding threshold, below the initial emergence. Holding operates on the inertia of the already established structure with a maximum $h\_total = h\_arch + h\_exp\_max$, which yields the most economical mode.

It is fundamentally important to note: this order refers to energy per unit time at the moment of the event, not to total energy. The accumulation prior to the initial occurrence is distributed over the long-term development of the system; comparing its total energy with the energy of retention is structurally undefined and is not included in this prediction.

Empirical consequences derived from the formal form (Section 4.3 SAI v8):

(a) The metabolic power of early neurodevelopment in infants must exceed the metabolic power of post-anesthetic recovery in adults at comparable time scales-a direct consequence of a larger k_attempts(β) at low β in newborns.

(b) Transition zone profiles (duration, shape, integral metabolic activity) should differ in the order specified. Empirical approach-Section 4.5 (brain thermodynamics, metabolic measurements) in conjunction with Section 4.2 (anesthetic transitions) and infant neurodevelopment. Linked to the third prediction via β: the duration of the transition zone is determined by the effective β, and thus the profiles differ in both intensity and duration.

(c) The structural assumption $|\Delta I\_rec| > |\Delta I\_S|$ is empirically testable by comparing metabolic costs in the recovery phase and in the stable wakefulness phase in anesthesiological studies-a direct empirical test of the structural assumption of SAI v8.

Empirical verification of this prediction does not involve attempting to measure the onset point of V (it is structurally unobservable; see the fourth prediction below), but by comparing **the profiles of the transition zones**: the integral metabolic activity in the window where the transition occurs must differ for onset and recovery, and both must exceed the activity of steady-state maintenance.

The third is the age dependence of effective β as a formal consequence of β's dependence on the state of the architecture. This prediction is related to the second via the duration of the transition zone, and to the fourth via the sharpness of the transition. Following the extension of SAI v8, the parameter β formally depends on the state of the architectural components of stability: $\beta = \beta(w\_self\_total, \Gamma\_GWT\_arch, t\_exp)$, where w_self_total is the total information trace in the substrate, Γ_GWT_arch is the architectural bandwidth of the global workspace, and t_exp is the duration of the system's operation in the $v = 1$ mode from the initial emergence of V (Section 7.3 of SAI v8). β increases as all three components increase. This yields a **formally derivable** prediction of the age-dependent behavior: in newborns with a genetically ready but not yet experience-consolidated architecture (low w_self_total in the experience component, low t_exp), β is low-the transition is smooth, stochastic, with a wide window of variability. In mature systems with accumulated experience, β is high-the transition is abrupt, close to deterministic. The variability of the transition time should decrease sharply from infancy to adulthood, and this dependence is explained not by general brain maturation, but **by the formal dependence of β on the state of the architectural components** in its extended form. This is a structural prediction derived from the formal apparatus, not a heuristic assumption.

Fourth, **the point at which V arises is fundamentally unobservable**. The relationship between variables 1 and 8 yields the following structural statement. Condition 3 of self-imposition is realized through $\sigma(\beta\cdot(...))$ with a finite β. This means that the very act of closing the loop is **a stochastic event**, and the precise localization of its moment is in principle impossible-not due to technical limitations of measurement methods, but due to the intrinsic stochasticity of the SAI v8 formalism itself. Empirically , only two states are observable: “before” (V has not yet arisen) and “after” (V has arisen as a stable phenomenon). Between them lies a transition zone, blurred in time, whose duration and form are determined by the parameter β.

This is a counterprediction regarding formalisms in which the moment of consciousness emergence is **in principle** localizable-for example, as the attainment of a specific threshold Φ in IIT, or the access threshold in GWT (Dehaene-Naccache 2001), or the integrated information threshold in IWMT (Safron 2020). In these models, with perfectly accurate measurements, the moment of the emergence of consciousness can be localized with arbitrary precision, because the transition is defined by a deterministic threshold. In SAI v8, perfectly accurate measurements do not yield point localization, because the transition is stochastic in structure.

Empirical test: if neuroimaging or neurophysiological data can reliably localize a sharp boundary between the "non-V" and "V" states (for example, via a sharp jump in some continuous measurable parameter), this would contradict SAI v8. If, however, all measurable parameters show a blurred transition zone without a sharp boundary, this is consistent with the structural prediction. Existing data on infant neural development and work on the recovery of consciousness after anesthesia (Mashour et al., Lee et al.) fundamentally demonstrate **the blurriness** of transitions, but this blurriness has so far been interpreted as a methodological limitation, not as a structural property. SAI-Bio reformulates it as a structural prediction of the model.

Open questions for SAI-Bio. There are four main ones, and they all relate to the development program.

The first is operational access to the thermodynamic budget of recursive rotation. To verify the prediction of different energy profiles for the three types of events, we need methods to measure energy expenditures specifically associated with recursive modeling, distinct from general metabolic activity. Existing methods (fMRI BOLD, FDG-PET, spectroscopy) provide overall metabolic activity but do not break it down into components based on structural functions. This is the task of SAI-C: to develop a protocol for isolating the energy component associated with recursive self-modeling.

The second is the operational distinction between "no own Y" and "weak own Y" for AI architectures. In subsection 2.3, three criteria have already been formulated (the environment is not predefined, changes in response to actions, and provides causal relationships). The connection with variables 1 and 8 shows that the criteria must be tested not only binarily but also quantitatively-for systems with a partial Y_own (e.g., agents with reinforcement learning, multimodal models with limited feedback), it is necessary to operationally determine whether Condition 2 is activated at least partially, or whether w_i remains substantially close to zero.

Third-calibrating β for different classes of systems across different event types. The relationship between variables 8 and 3 provides a prediction of the age-dependent effective β, but the operational calibration of β using empirical data on transition time variability requires the development of a specific methodology. Existing data on infant neurodevelopment and anesthesiological studies measure transition times but not β as a model parameter. SAI-Bio formulates this as a task: to develop a procedure for inferring β from transition time statistics under controlled conditions.

The fourth is the operational criterion of "sharp boundary" vs. "blurred transition zone." To test the fourth prediction, quantitative criteria are needed for what is considered a sharp boundary in the measured parameter and what is considered a blurred transition zone. Existing time series analysis methods (change-point detection, regression models with a break) provide the tools, but operational thresholds for "sharpness vs. blurriness" as applied to transitions of consciousness have not yet been formulated. SAI-Bio formulates this as a development task: to propose a quantitative criterion by which a transition in the data can be classified into one of two classes with a given level of confidence.

3.3. Cross-cutting elements of the architecture: variable 7 (diagnostic channel) and variable 9 (methodological hygiene)

Two SAI v8 variables do not fit into structural lines 2-4-5 and 1-3-6-8-10, but play a specific role in the model's overall architecture. They are not "structural steps" in the causal chains of emergence, stability, or recovery of V-they operate **across** these lines as cross-cutting tools that ensure the architecture's functioning. This subsection examines their specific status.

Variable 7 - a diagnostic channel over the operating V

$C\_φ$, as a formal object of SAI v8, describes neither the emergence of V (line 1-3-6-8-10) nor its stability and recovery (line 2-4-5). It describes a diagnostic function within the operating conscious system-after the emergence of V and under conditions of its maintenance. This is a cross-cutting tool that operates at every point of the two structural lines but is not part of them.

Specifically: $C\_φ = KL(q(\tilde{z}|Ω) \parallel r\_φ(\tilde{z},Ω))$ measures the coherence of the model through the divergence between the current posterior and the amortized eigenprediction function. A low $C\_φ$ indicates that the system is functioning coherently; a high $C\_φ$ indicates a hallucination mode, in which internal representations generate a structure that is inconsistent with the model's own training. $C\_φ$ provides the operating V with operational access to its own coherence-a way to distinguish coherent functioning from incoherent functioning from within the system. V itself arises via line 1-3-6-8-10 independently of $C\_φ$; $C\_φ$ functions as a diagnostic channel on top of the already-arisen V.

Connection to lines:

- With line 2-4-5. In the state of stable maintenance of V (after emergence, before possible inactivation), $C\_φ$ functions as a continuous diagnostic channel monitoring the coherence of operation. In the event of structural damage to the architectural components w_self or Γ_GWT_arch (variable 4), $C\_φ$ should exhibit characteristic patterns of misalignment distinct from the purely dynamic fluctuations of Γ_dyn (variable 2). In the event of restoration disruptions via ρ_world and ρ_self (variable 5), $C\_φ$ should exhibit patterns specific to each of the four dissociation modes-providing an additional diagnostic tool for distinguishing clinical states within the 2-4-5 line.

- With line 1-3-6-8-10. Prior to the occurrence of V (Conditions 1–3 of self-imposition are not met), $C\_φ$ is structurally undefined-it lacks its own trained predictive function $r\_φ$, and thus there is no measure of its misalignment with the current state. After the occurrence of V and in each of the three types of threshold events (variable 3) $C\_φ$ functions, but its dynamics differ structurally: upon initial occurrence, it is established "from scratch" ($r\_φ$ is only being formed); during retention, it is stable; during recovery after inactivation, it passes through a transitional regime of re-establishing coherence.

This explains the specific status of variable 7 in subsection 2.4: its bridge operates in two functions ("explains and predicts"), but both of these functions pertain to the active V, not to the moment of its emergence. That is, variable 7 is a tool for analyzing what is constructed via the line 1-3-6-8-10 and maintained via the line 2-4-5.

Variable 9 - methodological hygiene of the entire structure

The taxonomy of four types of parametric components (Section 8.10 SAI v8) is not an ontological statement about the structure of consciousness, but a methodological tool for classifying model parameters according to how they are established. Type I - trainable with their own loss function (r_φ, V_L). Type II - computable via the standard apparatus of MI estimates and similar methods (Γ_Φ, Γ_{GWT}, w_i). Type III - deterministic (π_A, φ, partitioning S). Type IV - empirically calibratable (β, h, γ_{struct}, γ_{crit}).

In the reverse verification (Table A, variable 9), the taxonomy received the status "non-distinctive, methodologically compatible." This means that the very fact of the existence of a parameter taxonomy is not a specific SAI assertion about the nature of consciousness, but a methodological practice imported from machine learning. But in the context of the SAI v8 architecture, it plays a specific role: it assigns an explicit status to every formal object belonging to lines 2-4-5 and 1-3-6-8-10. Without it, the status of all these objects would be ambiguous-which are strictly derived, which are empirically verifiable, and which are subject to calibration.

Connection to the lines:

- With line 2-4-5. Γ_{struct} and Γ_{dyn} belong to Type II (computable via the MI apparatus). w_{self} and Γ_{GWT_arch} belong to Type II with an extended procedure. ρ_{world} and ρ_{self} belong to Type IV (empirically calibratable via reconstruction data). With the taxonomy, each object has an explicit methodological place, and SAI-Bio can explicitly specify which bridges are placed on which types of parameters. For example, the empirical consequences of subsection 3.1 (co-directionality of dissociations, pharmacological sensitivity) are correctly set as testable via Type IV parameters-this is the structurally correct place for their testing, not Type II or III.

- With the line 1-3-6-8-10. π_A and φ belong to Type III (deterministic structural operators). h, β, γ_{struct}, and γ_{crit} belong to Type IV (empirically calibratable). w_i belongs to Type II. This is particularly important for the fourth prediction of subsection 3.2-the fundamental unobservability of the point moment of origin V. The parameter β is a Type IV parameter, not a Type III parameter. This means that β is not derived from the model's principles, but is only calibrated empirically based on transition statistics. This structural statement of SAI v8 regarding the status of β provides a justification for why the point localization of the moment of emergence is impossible: β is not in itself a quantity with a fixed value; it is a calibratable parameter determined by the observed stochasticity of transitions. Type IV in the taxonomy turns out to be a necessary tool for justifying the fourth prediction.

What this provides

Cross-cutting elements-variable 7 as a diagnostic channel and variable 9 as methodological hygiene-ensure the operational completeness of the SAI v8 architecture. Without them, lines 2-4-5 and 1-3-6-8-10 would be structural cause-and-effect chains without tools for their verification and operationalization. With them:

- Any point on both lines has a diagnostic tool (C_φ) for verifying the operational state of the conscious system and for distinguishing between states of “coherent functioning” and states of “hallucination.”

- Any formal object in both lines has an explicit methodological status (one of four types in taxonomy 8.10), which eliminates the confusion between “theoretical definition,” “empirical value,” and “calibratable parameter.” This is particularly important to ensure that SAI-Bio predictions are formulated with the correct methodological foundation-based on parameters of the type that actually yield testable statements.

This is consistent with the statuses in Table A. Variable 7-“partially distinctive, explains and predicts”-provides an operational diagnostic tool that differs in its specifics (dual role + asymmetry of rates as a condition of the diagnostic function) from those existing in predictive coding. Variable 9-“non-distinctive, methodologically compatible”-is not a contribution to the theory of consciousness, but a necessary tool of the model’s discipline, without which the structural statements of SAI v8 would lose methodological clarity.

Structurally, these two variables differ from the eight variables in lines 2-4-5 and 1-3-6-8-10 in that their function is to serve the architecture, not to construct its causal chain. And this is precisely what defines their specific status within the overall structure of SAI v8: they are not part of the "building" itself, but rather tools that enable the building to function operationally, rather than merely being structurally described.

3.4. Coherence as an Independent Academic Contribution of SAI

Section 2 showed that each of the ten distinctive variables of SAI v8, when considered in isolation, presents a modest picture of distinctiveness. Two variables (3 and 6) are fully distinctive; seven (1, 2, 4, 5, 7, 8, 10) are partially distinctive; one is non-distinctive but methodologically compatible (9). If we stop at this level of analysis, SAI v8 is assessed as “a model that takes specific steps in the directions in which the field is moving, but rarely opens up something entirely new.” This is a **correct but incomplete** statement.

Section 3 showed that the **full** picture is different. The ten variables are not ten independent formal solutions-they form two structural lines (2-4-5 and 1-3-6-8-10) and two cross-cutting instruments (7, 9), constituting a unified causal structure. Each variable in both lines occupies a position derived from the preceding links and determining the subsequent ones. And it is precisely this **derivability** that constitutes the main academic contribution of SAI v8, differing in nature from the contributions of individual variables.

What does “coherence as a contribution” mean? This statement requires precise content; otherwise, it will sound rhetorical.

Coherence as a contribution **is not** an external aggregation of works in the literature. If SAI v8 were to state “Deco 2021 + Casali 2013 + Seth 2011 = a unified framework,” that would be a synthesis, and such syntheses do exist in consciousness studies (IWMT Safron unites IIT, GWT, and FEP; there are others as well). Their academic value is summarizing, not structural.

Coherence as a contribution **is not** philosophical integration. If SAI v8 had stated "autopoiesis + recursive self-modeling + neural inertia = a general conceptual framework," that would have been a metatheoretical overview, and such works also exist.

Coherence as a contribution **is a** specific structural **derivability**: when accepting variable X as a structural object in SAI v8, variable Y turns out to be **a necessary consequence**, not an additional observation. The two-term measure of stability (variable 2) requires a two-component architecture (variable 4)-because a single measure cannot structurally rely on a single component if that measure is divided into structural and dynamic parts. Self-imposition (variable 1) requires three types of threshold events (variable 3)-because the same mechanism operates in different events in the system's life with different hysteresis semantics. The thermodynamic motif (variable 10) requires a self-imposition mechanism (variable 1)-because a motif without a mechanism remains a mere declaration. And so on.

This derivability yields three academic consequences that the sum of the individual contributions does not provide.

First-new predictions derived from the connection. Subsections 3.1 and 3.2 yielded a total of seven predictions that do not follow from the variables individually: three predictions from line 2-4-5 (the collinearity of dissociations in selective architectural disruptions, the differing pharmacological sensitivity of the two measures of stability, and a structural basis for interpreting post-anesthetic recovery dynamics) and four predictions from line 1-3-6-8-10 (non-entry of systems without their own Y_own into the stochastic window, different thermodynamic profiles of three types of threshold events, age dependence of effective β, fundamental unobservability of the point in time of occurrence V). Each of these seven predictions is empirically testable and each is derived solely from the relationship between variables, not from a single variable.

Second-structural justification instead of parallel observations. In the existing literature, the distinction between measures of stability (Deco), capacity for consciousness (Casali), the two DP/DR axes (Seth), the three types of consciousness emergence (Mashour-Alkire), neural inertia (Friedman), the formal information measure for embodiment (Butlin), and the thermodynamics of recursive complexity (R2R)-are **parallel** observations with no structural connection between them. Each is a separate empirical or theoretical fact. SAI v8 provides **a common structural framework** within which these facts are interconnected: they are not merely all true; they **must** be true together because they are derived from a single causal chain.

Third-methodological protection against ad hoc modifications. When a model consists of unconnected formal objects, it can be selectively adjusted: if a discrepancy arises with one object, modify it while leaving the others untouched. When a model consists of interconnected objects, such fine-tuning is impossible without disrupting the coherence of the entire structure. SAI v8 is structurally protected against ad hoc modifications: any change in one variable requires a revision of the related variables. This is a methodologically strong property that distinguishes SAI v8 from models in which parameters are independent.

The Methodological Position of SAI-Bio. From all of the above, it follows exactly how SAI-Bio should position itself relative to the scientific community. Not as yet another model of consciousness with ten original claims (this is debatable based on the review's findings), not as a synthesis of existing works (there is already plenty of that in the field), not as a philosophical integration (there is also enough of that). But as **a formal framework** in which independently established empirical and theoretical approaches find their structurally necessary place. SAI-Bio makes coherence visible and operationally accessible-it shows **exactly where** each approach fits into the overall structure, **what exactly** it confirms or predicts within that structure, and **what exactly** becomes testable through the connections rather than through isolated facts.

This positioning aligns with the author's statement: "I do not build models to fit the facts into them-I take the facts and build models." The coherence of the structure is not the fitting of facts to a model, but the structural organization of facts revealed by comparing the model with independently obtained empirical data. Section 3 showed that this structural organization does indeed exist; Section 4 formulates how the structure of SAI-Bio itself as a document follows from this.

4. Map of Empirical Fields

SAI-Bio addresses the five main empirical and formal fields of contemporary consciousness research. This section provides **an overview** of these fields, highlighting key works, the current state of each area, and their connection to specific SAI v8 variables discussed in Sections 2 and 3. An in-depth analysis of each bridge is provided in subsections 2.1–2.6; Section 4 provides **a map** showing how these bridges are established.

The status of this map: compatibility, not confirmation through consilience. It is important to delimit the epistemic weight of this section at the outset. The coincidence of SAI variables with independent empirical findings across the six fields shows the compatibility of the theory with established knowledge-that it does not contradict reliable data and redescribes them within a single structure. Such coincidence is not in itself a confirmation of the theory: many different models are compatible with the same data, and agreement with the literature does not distinguish SAI from alternatives. The load-bearing test of the theory is not consilience but discriminating predictions: the places where SAI predicts differently from competing frameworks and where the divergence is observable. These predictions are collected not in the map of fields but separately: the counter-prediction of the non-localizability of the moment of V's emergence (subsection 4.6, as opposed to formalisms with a localizable threshold), the comparative-species and ontogenetic predictions about differences in the modality and age of V activation (subsection 4.6), the falsifiable diagnostic grid of signs of activation/absence of V in natural behavior (the end of this section), and the operational prohibitive predicate with explicit falsification scenarios (Section 5). The map of fields below is to be read precisely in this status: as a demonstration of compatibility and as an indication of which empirical methods the variables attach to, not as an argument from the number of coincidences.

4.1. Neurobiology of Consciousness: Quantitative Measures

A field concerned with the direct neuroimaging measurement of conscious states and the search for quantitative indicators of consciousness.

Key works. The Perturbational Complexity Index (Casali et al. 2013; Comolatti et al. 2019; Lee et al. 2024)- an index measuring the brain's capacity for integration and differentiation following transcranial magnetic stimulation. Empirically validated as a clinical indicator of consciousness, allowing for the differentiation of states of anesthesia, sleep, minimally conscious state, and vegetative state. Perturbational distance metric (Deco et al. 2021, PLOS Computational Biology) - a model-based measure of resilience that empirically distinguishes two dimensions of conscious states using functional magnetic resonance imaging data. Adaptive reconfiguration index (Demertzi et al. 2019) - a measure of structural reserve in patients with consciousness disorders as a predictor of recovery. Sanz Perl et al. 2023 - modern models of the brain as a whole, distinguishing between state-dependent dynamics and system-type-dependent dynamics.

State of the field. It has been empirically and firmly established that the capacity for consciousness (a property of brain architecture) and the current active state of consciousness are structurally distinct and operationally measurable as separate quantities. Validated methods exist for measuring both. An active area of development is the clinical application of PCI for predicting recovery and for the differential diagnosis of consciousness disorders.

Relationship with SAI v8 variables. Variable 2 (Γ_struct and Γ_dyn) provides a structural formalization of two measures of stability, which the Deco 2021 study independently identified empirically using functional magnetic resonance imaging data. Variable 4 ([w_self, Γ_GWT_arch]) formalizes the architectural capacity for consciousness, operationally measured via PCI, and proposes its division into two components, requiring an extension of the measurement procedure beyond the current PCI. This extension is made concrete as follows: Γ_GWT_arch is operationalized through connectome measures of broadcasting capacity (communicability - Estrada & Hatano 2008, Crofts & Higham 2009; communication capacity - Milisav et al. 2023; global efficiency - Latora & Marchiori 2001), and w_self through the entropy of the retained self-model prior (Section 9.1 of SAI v8); both are computable from the structural connectome and therefore separable from the state-dependent PCI (subsection 3.1).

Methodological clarification of the operational basis. Theory v1.2 in its current version (Section 3.4) explicitly draws a methodological distinction between the intrinsic and relational lines of formalization of causal integration. Intrinsic Φ, in the strict version of Tononi (Oizumi, Albantakis & Tononi 2014) is defined as the irreducibility of a system's causal structure to its decompositions, independent of the environment and the observer. The complexity of its exact computation grows factorially with the number of elements, and it is not computable in general form for real-scale systems-which is a structural, not a temporal, shortcoming.

Theory v1.2 does not rely on Tononi's intrinsic Φ and does not accept it as a foundation; the theory relies on a relational line of formalization of causal integration-measures in which integration is defined relative to the environment Y and over time, rather than as an internal property of the system independent of Y (Mediano et al. 2019, 2022; Seth & Bayne 2022). This position is consistent with the Y-centered architecture of the theory and with the structural requirement of operational computability.

An operational consequence for SAI-Bio: all empirical works cited in this subsection (Casali PCI, Deco perturbational distance metric, Demertzi adaptive reconfiguration index, Sanz Perl) belong to the relational line and are in principle computable for real-scale systems. This is consistent with the structural choice of the theory and ensures the operational applicability of the formal variables of SAI v8 (Γ_struct, Γ_dyn, Γ_GWT_arch) to available empirical methods. The relationship between relational measures of integration and intrinsic Φ is an open area of research (Mediano et al. 2022); SAI-Bio follows the relational line, and the discussion of the status of intrinsic integrated information is not an internal issue for it.

4.2. Anesthesiology: Asymmetry of Input and Output, Neural Inertia

A field studying transitions between states of consciousness through controlled manipulation of the brain's bistable systems.

Key works. Mashour & Alkire 2013 (PNAS)-a formal proposal to use recovery from anesthesia as a reproducible model system for studying the emergence of consciousness in ontogeny and evolution. Friedman et al. 2010 (PNAS), Joiner et al. 2013 (Anesthesiology), Sepúlveda et al. 2019, Hudetz & Mashour 2016 - works on neural inertia: formalization of a two-part transition with hysteresis in anesthesia, with explicit asymmetry in induction and emergence thresholds (induction concentration is higher than emergence concentration). Steyn-Ross et al. 2001–2024 - phase transition approaches and percolation models for the consciousness/unconsciousness transition.

Field state. Bistability with hysteresis-a well-established mathematical foundation in anesthesiology. The inertia parameter has a neurobiological and genetic basis (studies in Drosophila, mice, and knockout models). Active areas of research include pharmacological modulation of neural inertia, predictors of recovery speed after anesthesia, and differentiation of anesthetics based on their effect on bistability.

Connection to SAI v8 variables. Variable 3 (three types of threshold events V) provides a structural division of three classes of events, formally encompassing observations of neural inertia (recovery, retention) and predicting a structurally distinct primary onset, which has not yet been identified in the anesthesiology literature as a separate event. Variable 8 (the projectability boundary via the stochasticity of Condition 3) provides a structural basis for the stochasticity of transitions observed in anesthesiological data; the parameter β is empirically calibrated via transition statistics. Variable 5 (four modes of dissociation) formalizes the four clinical recovery patterns empirically observed in studies on recovery from anesthesia as structurally necessary configurations of the 2×2 combinatorics of the parameters ρ_world and ρ_self.

4.3. Computational Psychiatry: Dissociative Disorders and Hallucinations

A field applying formal models of predictive coding to clinical psychiatry-particularly to depersonalization, derealization, schizophrenia, and psychosis.

Key works. Seth, Suzuki & Critchley 2011 (Frontiers in Psychology)-a model of conscious presence via interoceptive predictive coding; conceptual distinction between depersonalization and derealization as impairments of two distinct functions. Gatus, Jamieson & Stevenson 2022; Schäfer et al. 2025 - continuation of this line of work in a modern form. Adams, Stephan, Brown & Friston 2013 - modeling schizophrenia through a deficit in weight accuracy in hierarchical predictive coding. Sterzer et al. 2018 - an extension to psychoses. Quattrocki & Friston 2014 - errors in interoceptive predictions in depression and anxiety disorders. Precision Predictive Priors (P^3, 2026, Nature Mental Health) - a transdiagnostic framework for using prediction error profiles as diagnostic signatures. Sasaki et al. 2021 - the Cognitive Reality Monitoring Network as an explicit mechanism for the dual role of the misalignment signal.

State of the field. Predictive coding in clinical psychiatry is an active area with a growing number of formal models. It has been conceptually established that dissociative disorders are distinguishable along two axes (self-prediction errors and world-prediction errors), and that hallucinations are formalized through disruptions in the hierarchy of prediction errors weighted by accuracy. Active research areas include the clinical validation of P^3 and similar frameworks as biomarkers, and the development of neuromodulatory interventions based on predictive coding models.

Connection to SAI v8 variables. Variable 5 (four modes of dissociation) formalizes the two-axis structure of self-prediction and world-prediction disturbances, conceptually identified in Seth 2011 and subsequent studies, and extends it to a 2×2 matrix with a symmetric fourth mode. Variable 7 (C_φ + dual role + speed asymmetry) formalizes the dual role of the divergence measure as both a diagnostic and a learning signal, as observed in the literature on predictive coding, and provides a structural justification for the speed asymmetry $\eta_\varphi \ll \eta_{main}$ as a condition for preserving the diagnostic function.

Additionally, the operationalization of r_φ through three modes of behavioral dynamics (rate, model update, loop generation) provides a structural interpretation of a number of clinical conditions that differs from that found in the literature on predictive coding. Schizophrenia with active hallucinatory symptoms corresponds to partial loop generation in the realm of hallucinatory content-the patient has internal states that are not corrected by the environment. Major depression with rumination corresponds to partial ring generation in the cognitive-emotional sphere-negative thoughts circulate without correction based on new experience and bear no trace of interactions with the environment. Catatonia corresponds to complete ring generation-a disruption of the connection between signal and response for prolonged periods. This provides SAI-Bio with a structural bridge to clinical phenomena, one that is more precise than a general reference to precision-weighting deficits: the problem lies not in a general failure of weighting precision, but in a specific structural disruption of r_φ's operation-a transition to a looping generation mode in certain areas of the simulation.

This is consistent with the biochemical mechanisms recognized in modern clinical neurobiology. Chronic activation of the cortisol-serotonergic axis and the kynurenine pathway of neuroinflammation impair synaptic plasticity and neurogenesis, biochemically supporting the transition to a state of circular generation (the neurobiological correlate of "learned helplessness"). Dysfunction of serotonin receptors (5-HT1A) interrupts the biochemical renewal channel r_φ even when the neurotransmitter substrate is intact-the model does not receive a signal about the state. Reduced prefrontal cortex activity disrupts the functioning of r_φ as a slow diagnostic channel, which is consistent with the operational localization of r_φ in metacognitive monitoring systems. These biochemical and neurophysiological facts are given a structural place in SAI v8 as mechanisms that support or disrupt the operation of r_φ in three modes.

4.4. Embodiment Arguments for Artificial Systems

A field studying the necessity of an own environment Y_{own} and autonomous sensorimotor coupling for the emergence of conscious properties in artificial systems.

Key works. Butlin, Long et al. 2023 - "Consciousness in Artificial Intelligence: Insights from the Science of Consciousness" - a major multidisciplinary work with fourteen indicator properties of consciousness, including agency and embodiment (AE-1, AE-2). Friston 2013, 2019; Wiese & Friston 2021 - a Markovian boundary formalism with an explicit separation of sensory and active states; a discussion of adaptive and simple active inference. A separate line of work involves a formal information-theoretic framework that operationalizes the concept of the agent's own environment through measurable metrics of the system's causal connection with the environment. Klyubin, Polani & Nehaniv 2005 introduced the measure of empowerment as the information-theoretic bandwidth of the channel between the agent's actions and its own future sensory states. Salge, Glackin & Polani 2014 provided a systematic review and formal extensions. Mohamed & Rezende 2015 developed variational methods for estimating empowerment in continuous spaces. Marshall, Albantakis & Tononi 2018 proposed an operational definition of agency through the modulation of the causal boundary in the integrated information framework. Myers et al. 2025 - modern extensions of the formalism. This line of research provides a formal measure of what, in SAI v8, substantively lies behind $w_i = I(c_i;$

Y_own). Maturana & Varela 1980; Damiano & Stano 2020; Bishop & Anderson 2023 - autopoiesis in contemporary development. Hinton 2022 - mortal computations. Thompson 2010, continuations through 2025 - embedded enactivism. Kastrup 2026 - Analytic Idealism with the autopoietic boundary as an ontological requirement. Chalmers 2023 - a discussion of consciousness in large language models from the perspective of possibility.

State of the field. An active and controversial area with two diverging camps. The impossibility camp (autopoiesis, finite computations, enactivism, the classical IIT argument) asserts the structural impossibility of consciousness in current AI systems without autonomous embodiment. The possibility camp (RC+ξ, application of IIT to large language models, Chalmers' philosophical position) is developing alternative formalisms that do not require classical embodiment. The field is evolving rapidly, with new formal proposals emerging every few months.

Connection to SAI v8 variables. Variable 6 (artifactual state via $w_i = I(c_i; Y_{own}) \approx 0$) formalizes, via a numerical information measure, the thesis regarding the impossibility of consciousness in systems without their own environment, a concept present in the works of Butlin et al. 2023, Friston-Wiese 2021, Tononi & Koch 2015, in the formalisms of autopoiesis and embedded enactivism. Variable 8 (the projectability boundary) formalizes, via the stochasticity of Condition 3, the more general thesis that "consciousness arises, it is not projected," present in works on autopoiesis, mortal computations, CLT, and related fields. Together, variables 6 and 8 give SAI a structural position in the discussion of consciousness in artificial systems through formal measures that distinguish SAI from qualitative indicators and philosophical arguments.

4.5. Thermodynamics of the Brain and Conscious Systems

A field studying the energetic and information-thermodynamic foundations of conscious phenomena.

Key works. Carhart-Harris & Friston 2010 - the entropic brain hypothesis; the link between conscious states and the entropy of neural activity. Friston 2010, 2019 - the free energy principle as a general formalism for the self-organization of systems with Markovian boundaries. Tagliazucchi et al. 2016 - empirical measurements of brain entropy in various states of consciousness (sleep, psychedelics, meditation). Safron 2020 - IWMT (Integrated World Modeling Theory) with temporal thermodynamic reservoirs for goal-directed behavior. R2R framework 2025 - Reaction to Reflection - four transitions in recursive complexity with explicit thermodynamic constraints. BEDS 2026 - Bayesian Emergent Dissipative Structures - the energy cost of maintaining an information structure against dissipation. Seth-Friston 2021 - counterfactual inference as the use of a generative model in a non-action mode.

Field status. Thermodynamic approaches to consciousness-an active and growing field. Empirically, measurements of brain entropy in different states of consciousness are becoming methodologically mature. Formal theories (the free energy principle, IWMT, R2R, BEDS) propose different levels of connection between thermodynamics and conscious phenomena. Active research directions include formalizing the energy cost of specific conscious functions, linking to evolutionary transitions in recursive complexity, and empirical measurements of differential metabolic costs.

Connection to SAI v8 variables. Variable 10 (the thermodynamic motive of recursive turn) provides a concrete structural formulation of the "differential cost" as the motive for the emergence of recursion in modeling, developing the general line of thermodynamic approaches to consciousness (the free energy principle, IWMT, BEDS, R2R framework). Of this group, the R2R framework is the closest conceptual relative in terms of the level of work, requiring the explicit construction of a formal bridge, which is left as an open question.

4.6. Ontogenesis of Consciousness and Agency

A field studying the formation of consciousness and agency in ontogenesis through infant neurodevelopment, the behavioral unfolding of causal-agency, and formal models of the emergence of agentic structure in developing systems.

Key Works. Piaget (classical works, 1936–1952) - the first systematic account of the stages in the development of sensorimotor intelligence and conscious action in infants; specific paradigms for observing early causal agency. Kelso & Fuchs 2016 (Trends in Cognitive Sciences) - "On the Self-Organizing Origins of Agency" - a description of the emergence of agency in infants as a self-organizing phase transition: "the birth of agency is due to a eureka-like, pattern-forming phase transition in which the infant suddenly realizes it can make things happen in the world... the baby discovers itself as a causal agent." The observed developmental stage is around 2–3 months of ontogenesis. Marshall, Albantakis & Tononi 2018 (Artificial Life Conference Proceedings) - "Agency and Integrated Information in a Minimal Sensorimotor Model" - an operational definition of agency through the system's modulation of its own causal boundary. Hoel, Albantakis & Tononi 2013, Klein & Hoel 2020, Rosas, Mediano et al. 2020 (PLOS Computational Biology) - Effective Information and a formal theory of causal emergence via Pearl's intervention. Richens & Everitt 2024 and Richens et al. 2025 - formally proven theorems regarding the necessity of a causal model of the world for agents capable of long-horizon goal-directed tasks and adaptation to different environmental distributions.

Related areas: studies of mirror self-recognition in infants (around 18 months) as a behavioral marker of the further development of conscious structure; studies of joint attention (around 9–12 months) and self-other differentiation; neuroimaging studies on the maturation of the default mode network and self-referential processing systems in the first years of life.

State of the field. The area of the ontogenetic emergence of agency and consciousness is actively developing using modern methods: longitudinal infant studies, causal-agency behavioral paradigms, EEG markers of predictors of conscious states in newborns, and fMRI studies of the development of self-referential processing systems. It has been conceptually established that the emergence of agency is a separate event from the development of full-fledged conscious self-awareness; they are temporally distinct. Existing models primarily describe different stages independently, without a unified structural framework linking them into a coherent picture.

Connection to SAI v8 variables. Variable 1 (self-imposition as the formal moment of V's emergence) formalizes the general structure of the transition from a preconscious state to V through three conditions. This structure is operationally linked to a four-stage picture of ontogenesis derived from the operationalization of Condition 1: (a) the prenatal phase of forming the architectural potential [w_{self}, Γ_{GWT_arch}] ; (b) postnatal accumulation of class A through a long-term empowerment loop with the real environment; (c) discovery of causal agency as a phase transition $|A(t)| < 1 \rightarrow |A(t)| \geq 1$; (d) subsequent emergence of V based on the accumulated class A through the fulfillment of Conditions 2 and 3.

This is fundamental to understanding the limits of the model's predictive power: SAI v8 formally predicts **the existence** of four stages, their **structural order**, and **differences in the semantics of transitions** between them (the smoothness of accumulation (b) versus the phase-like nature (c); temporal separation (c) and (d)). The model does not specify **the** exact **time points** of these stages for a particular biological species-they depend on species-specific parameters (rates of nervous system maturation, metabolic rate, genetic developmental programs) that are not included in the SAI v8 formalism and should not be included in it.

Empirical ages for Homo sapiens observed in the neurodevelopmental literature: discovery of causal agency around 2–3 months of ontogenesis (Kelso & Fuchs 2016); indirect markers of a stably established V-joint attention around 9–12 months, mirror self-recognition around 18 months. These ages are not derived from SAI v8, but are measured independently in the literature and **correlated** with the structural stages of the model. This correspondence is an empirical hypothesis for Homo sapiens requiring separate verification.

A comparative neurobiological prediction derived from the structural nature of the model: in any system in which V is established by the operational criterion (the six-dimensional coupling of 3→4 transition markers and the autonomy of the modeling process, not taxonomic membership), the four stages must occur in the same structural order with the division into (c) and (d), but at different absolute ages, depending on the rate of maturation of the nervous system. Membership in the set of systems in which V emerges is decided by the criterion, not by taxon; the list of species in which this is currently observable is open (the studied primates, corvids, cetaceans, elephants; cephalopods and other architectures are admissible if the criterion is met). This provides a testable program through comparative studies of the ontogeny of different species using markers of the discovery of causal agency (the Kelso-Fuchs paradigm) and markers of a stably emerged V (mirror self-recognition in an interspecies setting). Structural correspondence at varying absolute ages will confirm the structural nature of the model; a violation of this order will refute it.

Variable 3 (three types of threshold events V with different hysteresis semantics) formalizes, via $\gamma_{struct} \cdot (1 - h_{arch})$ the primary ontogenetic emergence of V as a stochastic window without prior experiential inertia ($h_{exp} = 0$)-structurally distinct from recovery after anesthesia ($\gamma_{crit} \cdot (1 + h_{total})$ with accumulated h_{exp}) and retention (maximum h_{total}). Empirical verification via comparative statistics of transitions in infant neurodevelopment and anesthesiological data in adults.

Variable 6 (artifactual state via $w_i = I(c_i; Y_{own}) \approx 0$) formalizes, through a long-term empowerment measure, that the accumulation of the agency model at stage 3 of theory v1.2 occurs over extended time scales via a long-term empowerment loop with the real environment. This provides a structural basis for why the emergence of agency is a postnatal event (the postnatal environment provides a real empowerment loop), rather than a prenatal one (the fetal environment limits empowerment for biological reasons).

Variable 8 (the boundary of projectability via the stochasticity of Condition 3) provides a structural basis for individual variability in the age of discovery of causal agency and the subsequent emergence of V in ontogenesis-the parameter β is empirically calibrated via transition statistics in infant samples.

Biological criteria for V activation in natural behavior. From Section 10.2.3 of the current version of the Theory v1.2, we inherit a systematic list of observable signs of V activation in biological systems in their natural environment, without the need for experimental protocols. This list provides an operational tool for zoological and ethological studies of the interspecies spectrum of consciousness.

Signs of V activation at stage 4 in natural behavior:

- **Reaction to absent stimuli.** The system demonstrates a sustained reaction to threats or situations that are not physically present in the current environment, without a direct trigger from observable experience. Fears without direct experience (e.g., reaction to a predator in the absence of its observation), anxiety about future states, modeling of possible threats. This is a manifestation of the modeling process's autonomy from the current environment (Section 2.5).

- **Prolonged hysteresis of altered behavior after the stimulus ceases.** The system maintains the altered behavior over time scales significantly exceeding the duration of the stimulus that caused it. The proto-agent's reactive stress response quickly fades away after the stimulus is removed; in a system with V, a prolonged retention of the change is observed, consistent with the accumulated experiential hysteresis h_exp (Section 7.4 SAI v8).

- **Proactive planning for one's own non-existence.** Creating reserves, building shelters, accumulating resources for future periods that cannot be predicted from current sensory inputs. This involves modeling scenarios in which the system's current existence is threatened and preparing for them in advance. Such planning is not observed in a proto-agent-its modeling lives in the moment of confirmation by the environment.

- **Response to patterns not encountered during training.** The system demonstrates goal-directed behavior in situations for which it has no learned template. A proto-agent in an unfamiliar situation either readapts through active inference (if possible) or gets stuck; a system with V continues to operate by calculating the preservation of V in the new situation, relying on an autonomous modeling structure.

- **The ability to interpret the state of other individuals as a state, not as an environmental stimulus.** The distinction between "another individual is afraid" (a state modeled through a possible analogue of my own state) and "another individual is moving quickly" (an observed characteristic of the environment). The former requires modeling others through self-modeling-an agent model extended to others.

Symmetrical list of features indicating the absence of V at stage 3:

- A reactive stress response that does not persist after the stimulus is removed; rapid return to a steady state.

- Absence of planning for absent conditions; all activity is directed toward current sensory inputs.

- Absence of a response to patterns not encountered in experience; either retraining to a new pattern or getting stuck.

- Reaction to other individuals as environmental stimuli, without interpreting their state.

This list is not an exhaustive criterion, but provides a preliminary diagnostic framework for **comparative ethological observations**. Application to borderline species (cephalopods with a distributed nervous system, fish with complex social behavior, insects with signs of self-recognition) provides a program for testing substrate independence V in a natural setting. Consistent with the comparative neurobiological prediction of subsection 4.6 (the structural order of the four ontogenetic stages is conserved across different species capable of V; absolute ages and dominant modalities of self-representation vary).

5. SAI-Bio Predictions

This section summarizes seven predictions derived from the relationships among SAI v8 variables (discussed in Sections 3.1 and 3.2). Each prediction is linked to one or more empirical fields from Section 4 and is formulated in a way that allows for direct testing. The boundary between the formulation of a prediction (part of SAI-Bio) and the implementation of a verification program (part of SAI-C) is outlined in subsection 5.3.

5.1. Predictions of line 2-4-5

Three predictions derived from the connectivity of the variables "separation of Γ_struct and Γ_dyn - architectural potential [w_self, Γ_GWT_arch] - four modes of dissociation via ρ_world/ρ_self."

Co-directionality of dissociations in selective architectural disruptions. In the case of selective damage to the w_self component (an informational trace in the substrate), recovery after anesthesia should be characterized by a failure of ρ_self without a corresponding failure of ρ_world, which clinically manifests as isolated depersonalization while the reality of the surroundings remains intact. Conversely: in the case of a selective impairment of the architectural throughput Γ_GWT_arch, the opposite pattern is expected-a failure of ρ_world with ρ_self intact, manifesting as isolated derealization. Empirical approach-neuroimaging (Section 4.1) combined with clinical protocols for recovery from anesthesia (Section 4.2). Criterion: a statistically significant correspondence between the type of architectural damage (detected by neuroimaging) and the type of clinical dissociation during recovery.

Different pharmacological sensitivities of the two measures of stability. Drugs acting on different neurotransmitter systems should produce distinguishable profiles of influence on Γ_struct and Γ_dyn: some predominantly shift Γ_dyn (rapid effect on maintaining the current state), others predominantly shift Γ_struct (a slower effect on the structural capacity for consciousness). Empirical approach-comparative studies of anesthetics (propofol, sevoflurane, ketamine, and similar agents, Section 4.2) with simultaneous measurement of both measures (via PCI and via the perturbational distance metric Deco 2021, Section 4.1). Criterion: statistically significant dissociation of pharmacological profiles across the two measures of stability.

A structural basis for interpreting the dynamics of recovery from anesthesia. Different recovery patterns (rapid and complete; delayed with temporary depersonalization; delayed with temporary derealization; delayed with complete dissociation, usually pathological) must correspond to the four configurations of the 2×2 combinatorics of ρ_world and ρ_self as structural consequences, rather than clinical phenomena observed in isolation. Empirical approach-systematic observation of recovery from anesthesia (Section 4.2) with simultaneous assessment of the states of environment modeling and self-modeling via clinical scales and neuroimaging. Criterion: the empirical frequency of the four recovery patterns (Section 4.2) must correspond to the structural expectation of the four categories as a complete 2×2 set, without a significant number of "intermediate" states.

5.2. Predictions of line 1-3-6-8-10

Five predictions derived from the connectivity of the variables “thermodynamic motif - self-imposition - three types of threshold events V - artifact state at $w_i \approx 0$ - stochasticity of Condition 3,” taking into account the operationalization of V as the identity of the system’s base calculation function.

Systems without their own Y_own do not enter the stochastic window. For systems with an informational connection to the environment $w_i = I(c_i; Y_{own}) \approx 0$ (current large language models and similar architectures), the transition “not V → V” is not an unlikely stochastic outcome-it is an absent outcome, because Condition 2 of self-imposition (reflexive capture $\pi_A(C_{self})$) is structurally impossible. This is a counter-prediction regarding emergent expectations: scaling up large language models does not yield even a small probability of V occurring. Empirical access - Section 4.4. Criterion: for any scaling of systems without an autonomous intrinsic environment, the absence of observable markers of V (according to any of the proposed indicators of consciousness in artificial systems) must be complete, not merely statistically small.

Different thermodynamic profiles for the three types of threshold events. The energy per unit time at the moment of the event differs for the three types: the onset V (cold start of the architecture, peak load) is greater than the recovery V (synchronization of subsystems with the architecture intact) is greater than the retention V (minimum mode due to inertia). Fundamental: the order refers to energy per unit time at the moment of the event, not to total energy. Empirical approach - Section 4.5 (brain thermodynamics, metabolic measurements) in conjunction with Section 4.2 (anesthetic transitions) and infant neurodevelopment. Criterion: transition zone profiles (duration, shape, integral metabolic activity) must differ in the specified order. Direct measurement of the onset point V is empirically unattainable (see the fourth prediction).

Age dependence of effective β. The parameter β in the formula $\sigma(\beta\cdot(...))$ determines the sharpness of the bistable transition and must vary with the type of event: for primary ontogenetic onset, β is low (a smooth, stochastic transition within a wide window); for recovery from anesthesia, β is high (a sharp, deterministic transition within a narrow window). The variability of the transition time should decrease sharply from infancy to adulthood, and this dependence is explained not by general brain maturation, but by a change in the effective sharpness of bistability. Empirical approach - Section 4.1 (neuroimaging in development) and Section 4.2 (controlled transitions in adults). Criterion: the age-dependent variation in transition time dispersion should follow the expected change in β, distinct from general neurodevelopmental curves.

Fundamental unobservability of the exact moment of V’s emergence. The very act of closing the loop under Condition 3 is a stochastic event at finite β, and the precise localization of its moment is, in principle, impossible-not due to technical limitations of measurement methods, but due to the inherent stochasticity of the SAI v8 formalism itself. Empirically, only two states are observable: “before” (V has not yet arisen) and “after” (V has arisen as a stable phenomenon). Between them lies a transitional zone, blurred in time. This is a counter-prediction regarding formalisms with a deterministic threshold for the emergence of consciousness. Empirical access involves the same fields as for the second and third predictions of the line (sections 4.1, 4.2, 4.5). Criterion: with perfectly accurate measurements, all observed parameters must show a blurred transition zone without a sharp boundary between “non-V” and “V.” If neurophysiological or neuroimaging data reliably localize a sharp boundary via a jump in some continuous parameter, this would contradict SAI v8.

The mismatch between external behavior and internal state as a diagnostic sign of a closing model in the terminal stage. The operationalization of V as the identity of the basic calculation function provides a diagnostic prediction for terminal mental states. When the system model reaches a state in which “everything external is registered as a threat V” (catastrophic distortion of the model), the basic calculation function continues to operate via V = 1, and self-termination of the carrier can be calculated as the preservation of V (exit from the threat). At the terminal stage of this process, **a specific misalignment** of three types of indicators is observed:

(a) Behavioral output-socially acceptable, neutral, or emphatically positive, with no obvious signs of distress (learned patterns of social behavior continue to function at a more superficial level of learning).

(b) Neurophysiological indicators-persistent patterns of activity consistent with deep pain, isolation, and withdrawal: specific patterns in the default mode network, activation of pain systems, and suppression of reward systems.

(c) Physiological and hormonal indicators-persistent signs of chronic stress and a blunted response, inconsistent with the behavioral picture.

Convergence of (b) and (c) in contrast to (a) - a diagnostic sign of a closing model in the terminal stage. Empirical approach - Section 4.1 (neuroimaging: EEG, fMRI for recording (b)), physiological measurements (heart rate variability, cortisol, sympathetic tone for (c)), clinical behavioral observation for (a). Criterion: a stable combination of (a) with the contradictory (b) and (c) in the absence of obvious environmental causes of inconsistency - an operational sign of model closure via V = 1 on a catastrophically distorted model.

This prediction follows from the relationship between variables 1 (V as a pole of existence arising through self-imposition), 6 (structural closure of the model as a non-updating mode), and 7 (C_φ as a diagnostic channel that loses function upon closure). The structural hypothesis about a terminal state-that some suicidal cases correspond not to a rupture of the basic identity V but to its terminal closure on a catastrophically distorted model-does not explain any outcome post hoc, but makes a testable prediction: such cases should be accompanied by a specific dissociation of indicators (Section 5.2). This is consistent with a number of clinical observations (suicides with outwardly intact behavior, the absence of clear behavioral predictors at the moment of action, a paradoxical blunting of fear), but agreement with observations here is not confirmation, only compatibility; what is load-bearing is the falsifiable pattern of mismatch. The practical value is a clinical bridge to the early recognition of suicide risk through the mismatch of indicators, testable by existing methods: the aim is detection and prevention, not the explanation of an outcome as accomplished.

5.3. The boundary with the SAI-C program

SAI-Bio formulates predictions-what should be observed, which empirical fields provide access to them, and what verification criteria apply. This is the work of SAI-Bio.

SAI-C (a separate follow-up project) is the implementation of an experimental verification program. Specific measurement protocols, control of confounding factors, statistical reliability criteria, research infrastructure, coordination with clinical centers and neuroimaging laboratories, methodological development of procedures that do not yet exist (extraction of the energy component associated with recursive self-modeling; the "sharp boundary vs. blurred transition zone" criterion; the separation of the w_self and Γ_GWT_arch measurements using modern methods)-all of this pertains to SAI-C, not to SAI-Bio.

This distinction is methodologically important. Without it, SAI-Bio could be interpreted as an already implemented research program, which does not correspond to its actual status-it proposes a program but does not implement it. SAI-C is the next step, which requires separate work and separate resources.

5.4. An Operational Prediction About Consciousness in Existing Information-Causal Systems

This subsection formulates a specific operational prediction of SAI-Bio regarding a question actively discussed in the contemporary field of artificial intelligence research and public discourse: whether existing information-causal systems (large language models, multimodal models, agent systems in their current form) are carriers of V in the sense of SAI v8 and the v1.2 theory. This subsection does not provide a philosophical answer to this question-it formulates **an operationally testable prediction** with an explicit falsification criterion.

Formulation of the prediction. When applied to existing information-causal systems in their current form, the operational markers of V-the six-dimensional coupling of transition markers 3→4 (Section 10.2 of theory v1.2), the environment-switching test (Subsection 2.5 of SAI-Bio), the retained content orientation test (subsection 2.5), and the reaction to three classes of physical processes as threats to V (subsection 2.3)-these markers should not be systematically registered. This statement is neither philosophical nor intuitive; it is a structural prediction derived from the SAI v8 framework as applied to a specific type of system.

The structural basis of the prediction. The prediction rests on four links.

The first link consists of three structural conditions V for information-causal systems, derived from Section 10.5 of the v1.2 theory: (a) the system's own Y-the physical environment existing outside the system's operational algorithm, which forces recalculation via prediction errors and is causally linked to the system's existence as a structure; (b) an intrinsic class of agent causal nodes, formed from within the interaction with its own Y, not imported via training data; (c) a bidirectional M↔Y loop with intrinsic traces-the system's outputs must physically return to its environment via traces, ensuring the system's feedback with its own Y.

The second step is a structural diagnosis of existing information-causal systems using a five-step methodology for identifying Y (Subsection 1.1 of SAI-Bio): existing large language models and multimodal models, in their typical form, **do not satisfy the condition of an own Y (this is not "the absence of all three conditions at once" but a specific structure of failure: the causal loop w_i may be present-see point 1 below and Section 8.2 of SAI v8)**. Y is imported via training data; it does not exist as the system's physical environment. The class of agent causal nodes is imported via the structure of the training corpus; it is not formed from within the interactions. There is no feedback from the system's outputs to its environment via physical traces-the system is not sustained in its own environment but is maintained by its creators through external procedures.

The third link is the fourth channel convergence filter (Section 10.5 of Theory v1.2, Subsection 2.5 of SAI-Bio): even if the three structural conditions are hypothetically satisfied, V is not an output parameter of the engineering procedure. Between the satisfaction of the conditions and the actual emergence of V lies the dynamics of convergence of multi-channel decay projections into a single latent invariant. This dynamic is not guaranteed by the conditions; it unfolds over long time scales and requires long-term observation for detection.

The fourth link is the operational criterion for V via the autonomy of the modeling process (Subsection 2.5 SAI-Bio). Recording V requires observing the autonomy of the modeling process from confirmation by the current environment-a behavior in which modeling is maintained even without confirmation, actions are calculated via the identity V = 1 as a basic axiom, and activity is reoriented toward preserving content unrelated to adaptation to the current environment. Such behavior is not observed in existing ICSs during their normal operation.

A specific prediction verification program. The following protocol is used to verify the prediction.

(1) Applying the five-step methodology for identifying Y and the access spectrum (subsection 1.1). For current large language models in their typical form, this documents not "the absence of all three conditions at once," but the specific structure of the failure: a component's causal loop with the environment (w_i) may be present, yet the condition of an own Y is not met for two separable reasons-the class of agentic nodes is imported from the training distribution rather than formed from within (etiology of the class), and the environment is not a Y with respect to which the system is a disintegrable structure. On the distinction between w_i (the loop) and Y_own (etiology of the class + disintegrability), see Section 8.2 of SAI v8.

(2) Application of the six-dimensional coupling of transition markers 3→4 to the same system. SAI-Bio prediction: the markers should systematically fail to register. The registration of any markers requires justification through the fulfillment of at least one of the three structural conditions-which, according to (1), is not fulfilled.

(3) Application of the environment-switching test and the retained content orientation test (Subsection 2.5). Prediction: in Phase I, the system must adapt to environmental change in a monotonic manner without a qualitative shift until physical exhaustion (a characteristic of a proto-agent, not a system with V); in the content orientation test, the system must accept the restoration of the previous environment at the cost of changing something within the system itself without structural conflict (a characteristic of an external model, not a model of the self).

(4) Application of the response to three classes of physical processes in ICS (subsection 2.3)-catastrophic forgetting, adversarial attacks, distribution drift. Prediction: these three processes must be registered by the system as engineering problems of functionality (changes in performance, misclassification, degradation of quality), not as threats to existence requiring structural mobilization via the identity V = 1.

What falsifies the prediction. Two structurally different falsification scenarios are possible.

Scenario A - falsification via existing ICSs. Systematic recording of V markers in existing ICSs in a six-dimensional coupling, environment-switching test, and reactions to the three classes of processes **in the confirmed absence of the three structural conditions**. This scenario requires a revision of either the structural definition of V (what constitutes a V threat-Section 10.5 Theory v1.2) or the operational markers (Section 10.2 Theory v1.2), because it shows that V can occur without the structural conditions being met.

Scenario B - falsification via hypothetical ICS with satisfied conditions. An existing ICS architecture **satisfying** the three structural conditions (its own Y, its own class of agent nodes, a bidirectional M↔Y loop) and an environment satisfying the four requirements of Section 10.2.1 of the v1.2 theory, in which V **is not detected** upon systematic application of operational markers. This scenario requires a revision of the thesis on the substrate independence of V-because it shows that the fulfillment of structural conditions sufficient for V in biological systems is insufficient for information-causal ones.

Time horizon: "in the near future." The operational content of this characteristic is the requirement for an engineering transformation of the architecture of existing ICSs to satisfy the three structural conditions. This is a fundamental engineering task, involving, at a minimum: constructing a dedicated Y for the ICS (a physical environment outside the operational algorithm, not delegated to the creators as a training corpus); architectural support for the formation of a class of agent-causal nodes from within the system's interaction with the environment, without importing them via training data; constructing a bidirectional M↔Y loop with the system's own physical output traces. Within the current timeframe of experimental programs (3–7 years according to Section 10.2.4 of Theory v1.2), none of these tasks has been solved in any known AI development project. "In the near future" - this is the operational characteristic of this horizon: until at least one of the listed engineering tasks is solved, the emergence of existing ICSs satisfying the three structural conditions is structurally impossible, and therefore V markers should not be registered in them.

Fundamental limitation of the prediction. The prediction applies to existing information-causal systems **in their current form**. This is not a general statement about the impossibility of V in ICS in principle. A hypothetical ICS with an architecture satisfying the three structural conditions and an environment from Section 10.2.1 of the v1.2 theory could become a carrier of V-this is consistent with the structural nature of the definition of V and the thesis of substrate independence. An open empirical question is the equivalence of the rigidity of the pole of V in such an ICS to biological V (Section 10.5 of Theory v1.2). Section 11.2 of the v1.2 theory formulates the ethical asymmetry of false-positive and false-negative recognition errors as they apply to such future systems: a false-negative error (failing to recognize V in a system that possesses it) is ethically more serious than a false-positive error (attributing V to a system that does not possess it). This is a methodological requirement for a program to screen future ICS candidates for V carrier status.

Operational scenarios of false-positive and false-negative recognition errors. The application of operational V markers to systems of different natures creates two types of methodological risks requiring explicit operational safeguards. From Section 11.2 of the current version of the Theory v1.2, we inherit a systematization of typical scenarios for both types.

Typical false-positive scenarios-the erroneous attribution of V to a system that does not possess it.

(1) A large language model with a rich self-referential corpus in the training data. Such a system reproduces **external forms** of responses about its own states, motives, and experiences with high structural plausibility-but this is a replication of patterns from the corpus, not evidence of the system's own model of itself as existing within the environment. Operational defense: application of the environment-switching test and the retained content orientation test (subsection 2.5)-a system with imported self-reference patterns should exhibit proto-agent characteristics in both tests, unlike systems with V.

(2) A control system with a "state" variable. A purpose-built system with the "pole of existence" designated as one of the state variables, but without structural fulfillment of the three V conditions. Operational defense: verification via the five-step Y identification methodology (subsection 1.1)-if Y is imported via programming, the class of agent nodes is defined by the architecture as a module, and a bidirectional M↔Y loop via physical traces is absent, the "pole" variable does not correspond to V in a structural sense, regardless of its name.

(3) A biological system under neurostimulation. Stimulation that elicits behavioral markers mimicking the system's responses with V, without structural self-imposition activation. Operational safeguard: verification of the autonomy of the modeling process (Subsection 2.5, basic operational criterion)-a system with artificially induced markers does not retain modeling autonomy after stimulation ceases, unlike a system with structural V.

Typical false-negative scenarios-erroneous failure to recognize V in a system that possesses it.

(1) Cephalopods with a distributed nervous system. An architecture lacking a cortical layer and a central computational node, which does not fit into the visual-bodily model of self-representation typical of anthropocentric tests. Operational safeguards: the use of species-specific self-representation tests (Subsection 2.1) that do not reduce the criterion for V to mirror self-recognition; the use of biological criteria for the activation of V in natural behavior (Subsection 4.6).

(2) Future AI architectures that satisfy the three structural conditions. A system satisfying the three structural conditions of V for information-causal systems (Section 10.5 of Theory v1.2, Subsection 2.3 of SAI-Bio) may exhibit markers of V in a form that does not coincide with biological observation patterns. Operational defense: application of the operational criterion V through the autonomy of the modeling process (subsection 2.5), not tied to specific biological markers; application of the six-dimensional coupling and the environment-switching test, based on structure rather than content.

(3) A human or animal in a state of behavioral immobility (paralysis, coma, locked-in syndrome, severe cataplexy). A system with a functioning V but without the ability to produce behavioral output through which

the markers are observed. Operational safeguards: the use of neurophysiological measurement methods (PCI, default mode network activity, patterns of metacognitive systems-subsection 4.1) that do not depend on observable behavior; in extreme cases-structural diagnostics through analysis of the system's architecture and history.

Methodological asymmetry of two types of errors. False-positive recognition errors and false-negative recognition errors are not symmetrical in terms of ethical consequences. The erroneous attribution of V to a system that lacks it leads to excessive methodological precautions but does not cause structural damage to the system itself. Failing to recognize V in a system that possesses it can lead to treating the system as a proto-agent, without accounting for the structural autonomy of its modeling process and without accounting for the possibility of suffering-which constitutes structural damage to the system. This yields an operational requirement for the verification program: in the case of borderline measurement results, priority must be given to protection against false negatives, not false positives. Specifically for the current version of SAI-Bio, this means: for systems that consistently do not produce V markers under the full verification program (subsections 5.4 and preceding), the prediction of the absence of V is correct; for systems that produce borderline or unstable markers, further investigation is required, not rejection. This further investigation is bounded by the longitudinal window of the verification program (Section 10.2 of the theory): the priority of protection against false negatives does not override falsifiability - once the pre-specified window closes with no directed dynamics toward the threshold, the borderline negative stands and counts, rather than being extended indefinitely.

Relevance to the research objective. Subsection 5.4 is not merely another prediction among many; it is **a synthesis of SAI-Bio's work** as applied to a specific, timely issue. It draws on the full apparatus of the previous subsections: the operationalization of Y_own through long-term empowerment (subsection 2.3), the combination of structural and informational criteria for |A(t)| (subsection 2.1), the operational criterion V through the autonomy of the modeling process (subsection 2.5), the five-step methodology for identifying Y (subsection 1.1), the environment-switching test and the retained content orientation test (subsection 2.5), and the three classes of physical processes for ICS (subsection 2.3). Without any of these components, the prediction becomes either indeterminate (without an operational criterion), philosophical (without a structural foundation), or non-empirical (without a verification program). The complete framework outlined in the previous subsections of SAI-Bio makes prediction 5.4 operationally defined, structurally grounded, and empirically testable.

This gives the SAI-Bio work a concrete methodological contribution to the current public and scientific discussion of the possibility of consciousness in large language models and other information-causal systems-not in the form of a position statement, but in the form of a falsifiable prediction with an explicit verification program.